Technical Training

Product information.

G12 PHEV High-voltage Battery Unit

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BMW Group University

Technical Training

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BMW Service

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:

⚠

Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

Contact: conceptinfo@bmw.de

The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **April 2016**
Technical Training

G12 PHEV High-voltage Battery Unit
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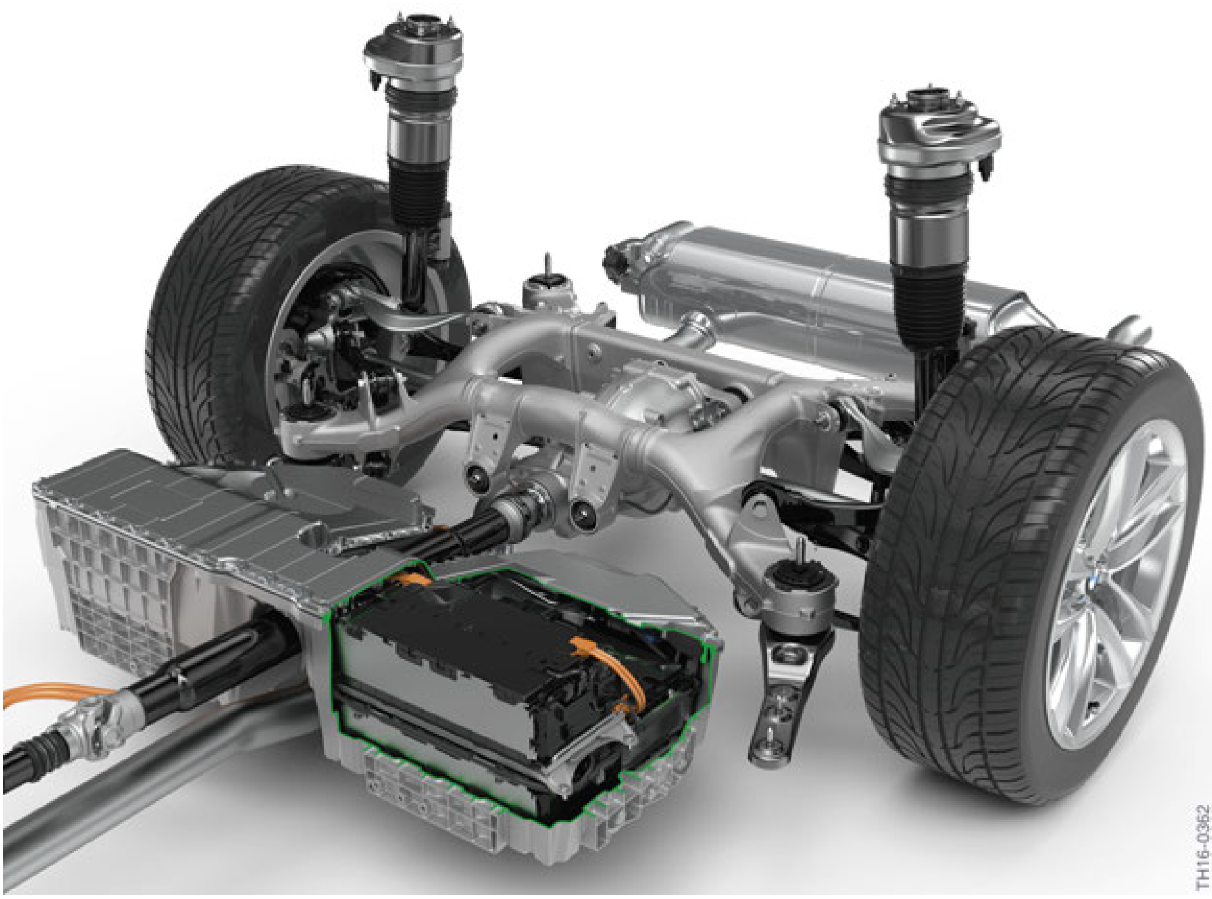
G12 PHEV High-voltage Battery Unit

1. Introduction

This Technical Training Manual describes the design of the high-voltage battery unit in the BMW 740e with the development code G12 PHEV and the special features relating to its repair. This document is not a replacement for the repair instructions, but should provide the reader with the necessary background knowledge and supplementary notes.

It is also possible for the G12 Generation 3.0 BMW hybrid vehicle to have components **inside** the high-voltage battery unit exchanged by qualified service technicians in order to repair the battery unit. The high-voltage battery unit is not available for the G12 PHEV as a complete replacement part. Special certification of the battery is offered to BMW dealership technicians for the repair of the high-voltage battery unit.



G12 PHEV high-voltage battery unit

G12 PHEV High-voltage Battery Unit

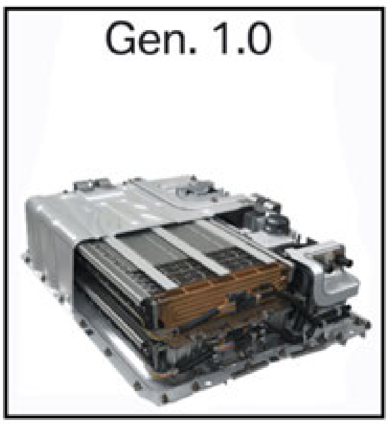
2. High-voltage Battery Unit

2.1. Overview

As in BMW ActiveHybrid motor vehicles the high-voltage battery unit is the energy storage device for the electrical drivetrain. The **G12 PHEV**, is the fourth **P**lug-in-**H**ybrid-**E**lectrical-**V**ehicle of the BMW brand after the F15 PHEV and F30 PHEV, The G12 PHEV is also equipped with the new generation Gen. 3.0 high-voltage battery unit. In case of potential defects, the Gen. 3.0 high-voltage battery unit can be repaired, in contrast to the Gen. 1.0 to Gen. 2.0, which needed to be completely replaced.

2.1.1. High-Voltage Battery Generations

Generation 1.0 to 2.0



Technical data	Gen. 1.0	Gen. 1.5	Gen. 2.0
Use	E72	F04	F01/F02H, F10H, F30H
Manufacturer	Bosch	TEMIC	BMW
Technology	Nickel metal hydride	Lithium-ion	Lithium-ion
Number of battery cells	260	35	96
Cell voltage Capacity	1.2 V 7.7 Ah	3.6 V 6.5 Ah	3.3 V 4 Ah
Nominal voltage	312 V	126 V	317 V
Voltage range	234 - 422 V	n.n.	n.n.
Storable energy capacity	2.4 kWh	0.8 kWh	1.35 kWh
Energy available for consumption	1.4 kWh	0.4 kWh	0.6 kWh
Max. output	57 kW. briefly	19 kW	43 kW
Weight	83 kg (182 lbs)	28 kg (61 lbs)	46 kg (101 lbs)

G12 PHEV High-voltage Battery Unit

2. High-voltage Battery Unit

Generation 3.0

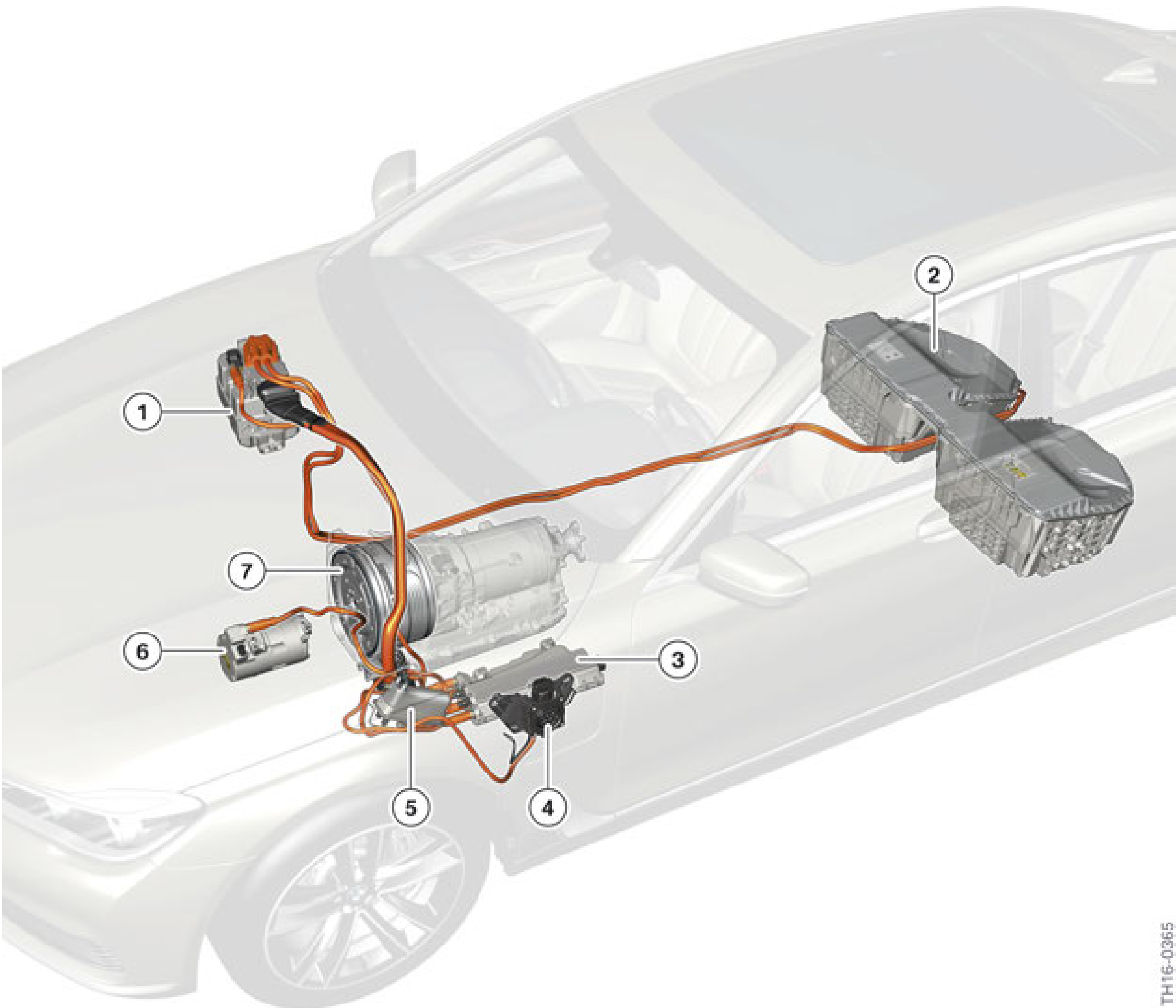
Technical data	**F15 Plug-in Hybrid Electric Vehicle**	**F30 PHEV**
Manufacturer	BMW	BMW
Technology	Lithium-ion	Lithium-ion
Number of battery cells	96	80
Cell voltage Capacity	3.7 V 26 Ah	3.66 V 26 Ah
Nominal voltage	355 V	293 V
Voltage range	269 - 399 V	225 to 328 V
Storable energy capacity	9.2 kWh	7.8 kWh
Energy available for consumption	6.8 kWh	5.8 kWh
Max. output	83 kW, briefly 43 kW continuous	65 kW, briefly 45 kW continuous
Weight	105 kg (231 lbs)	88 kg (194 lbs)

Like all BMW ActiveHybrid vehicles, the G12 PHEV has an electrical machine which can be used to charge the high-voltage battery. In addition to charging with the electrical machine, it is possible to charge the high-voltage battery unit of a Plug-in Hybrid Electric Vehicle externally from the domestic grid using a charging socket. The high-voltage battery can also be charged by brake energy regeneration.

The high-voltage battery of a vehicle with an electric motor is the equivalent to the fuel tank in a vehicle with a combustion engine. The high-voltage battery unit is installed centrally in front of the rear axle.

G12 PHEV High-voltage Battery Unit

2. High-voltage Battery Unit



G12 PHEV High-Voltage Components and High-Voltage Cables

Index	Explanation
1	Electric Motor Electronics (EME)
2	High-voltage battery unit
3	Convenience charging electronics (KLE)
4	Charging socket
5	Electrical heating
6	Electric A/C compressor (EKK)
7	Electric motor

2. High-voltage Battery Unit

2.1.2. Technical data

The high-voltage battery unit of the G12 PHEV consists of the following main components:

- Cell modules with the actual cells
- Cell Supervision Circuits
- Safety box
- Control unit for battery management electronics (SME)
- Four-part evaporator
- Wiring harnesses
- Connections (electrical system, refrigerant, venting)
- Housing and fastening parts

The battery cells are supplied by the Korean company Samsung SDI to the BMW plant in Dingolfing. This is where the cell modules are built from the battery cells and assembled along with the components to create a complete high-voltage battery unit.

The battery cells used in the high-voltage battery of the G12 PHEV are lithium-ion cells (cell type NMCo/LMO mixture). The positive terminals of the lithium-ion batteries basically consist of Lithium metal oxide. The designation NMCo/LMO mixture refers to the metals used for this cell type. It is a mix of nickel, manganese and cobalt on the one hand, and lithium manganese oxide on the other hand.

Graphite is used – as is the normal case – for the cathode. The lithium ions are deposited in the cathode during discharging. It was possible to optimize the characteristics of the high-voltage battery for use in an electric vehicle through this selection (high energy density, high cycle number). As a result of the materials used, the nominal voltage of the battery cells is **3.66 V**.

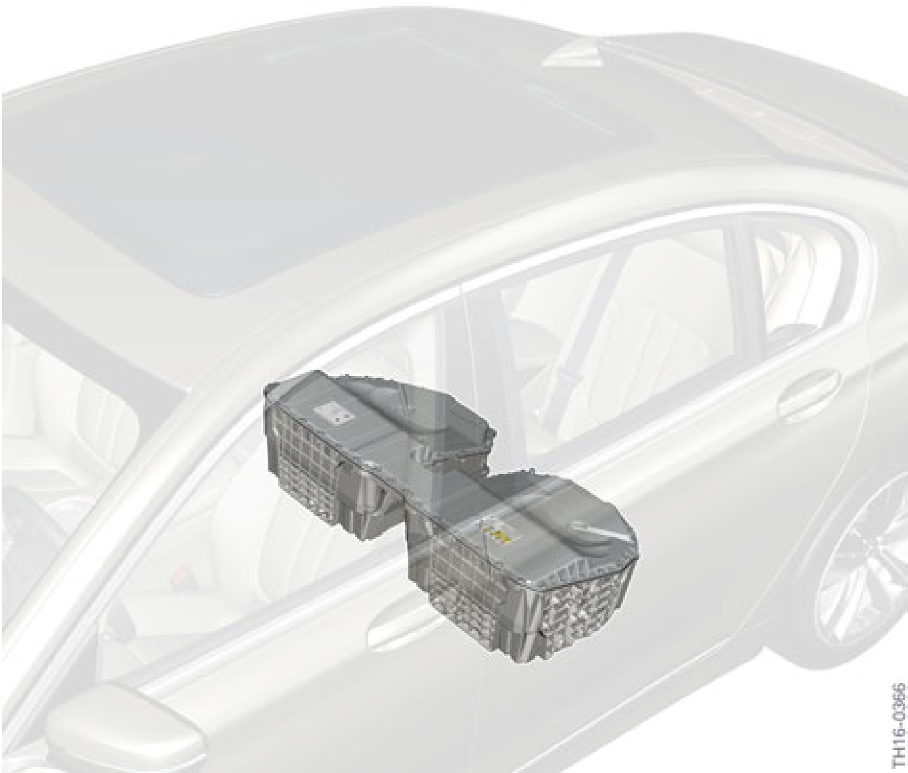
The following table summarizes some key technical data of the high-voltage battery in the G12 PHEV.

Voltage	351.4 V (nominal voltage) Min. 269 V – Max. 398 V (voltage range)
Battery cells	96 battery cells in series (each 3.66 V and 26 Ah)
Max. storable amount of energy Max. usable energy	9.2 kWh 7.4 kWh
Max. power (discharge)	83 kW (short-term)
Maximum power (AC charging)	3.7 kW
Total weight	112.6 kg (248 lbs) (without retaining brackets)
Dimensions	541 mm x 1134 mm x 271 mm
Cooling system	R1234yf

2. High-voltage Battery Unit

2.1.3. Installation location

The high-voltage battery unit is installed centrally in front of the rear axle. This has the advantage of lowering the G12 PHEV's center of gravity, which in turn results in improved driving characteristics. All connections are accessible from the underbody of the vehicle .



G12 PHEV, installation location of the high-voltage battery unit

The most important external features are:

- High-voltage cable and connection
- Interface for the 12 V vehicle electrical system
- Refrigerant lines or connections
- Labels
- Venting unit

The high-voltage battery unit also has an interface for the 12 V vehicle electrical system, in addition to the high-voltage connection. The control units integrated in the high-voltage battery unit are supplied with voltage, data bus, sensor and monitoring signals via this interface. The refrigerant circuit for cooling the high-voltage battery is incorporated in the battery.

The labels on the high-voltage battery unit inform people working with these components about the technology used and possible electrical and chemical dangers.

G12 PHEV High-voltage Battery Unit
2. High-voltage Battery Unit

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The electrical voltage of the high-voltage battery unit is well over 75 V. This is why **before** any work at the high-voltage battery unit the **electrical safety rules** must be observed:

1 Disconnect the system from the power supply.
2 Provide a safeguard to prevent unintentional restarting.
3 Verify that the system is isolated from the power supply.

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If the de-energized state cannot be clearly identified in the instrument cluster, further work at the vehicle is not allowed! **There is a danger to life!** The de-energized state must then be confirmed by a BMW qualified electrician using the appropriate measuring devices/measuring procedures. **In these cases, Technical Support must be contacted!** Furthermore, the vehicle must be locked and restricted using barrier tape.

The electrical lines (high-voltage and interface for the 12 V vehicle electrical system), as well as the refrigerant lines, can be disconnected without having to remove the high-voltage battery unit.

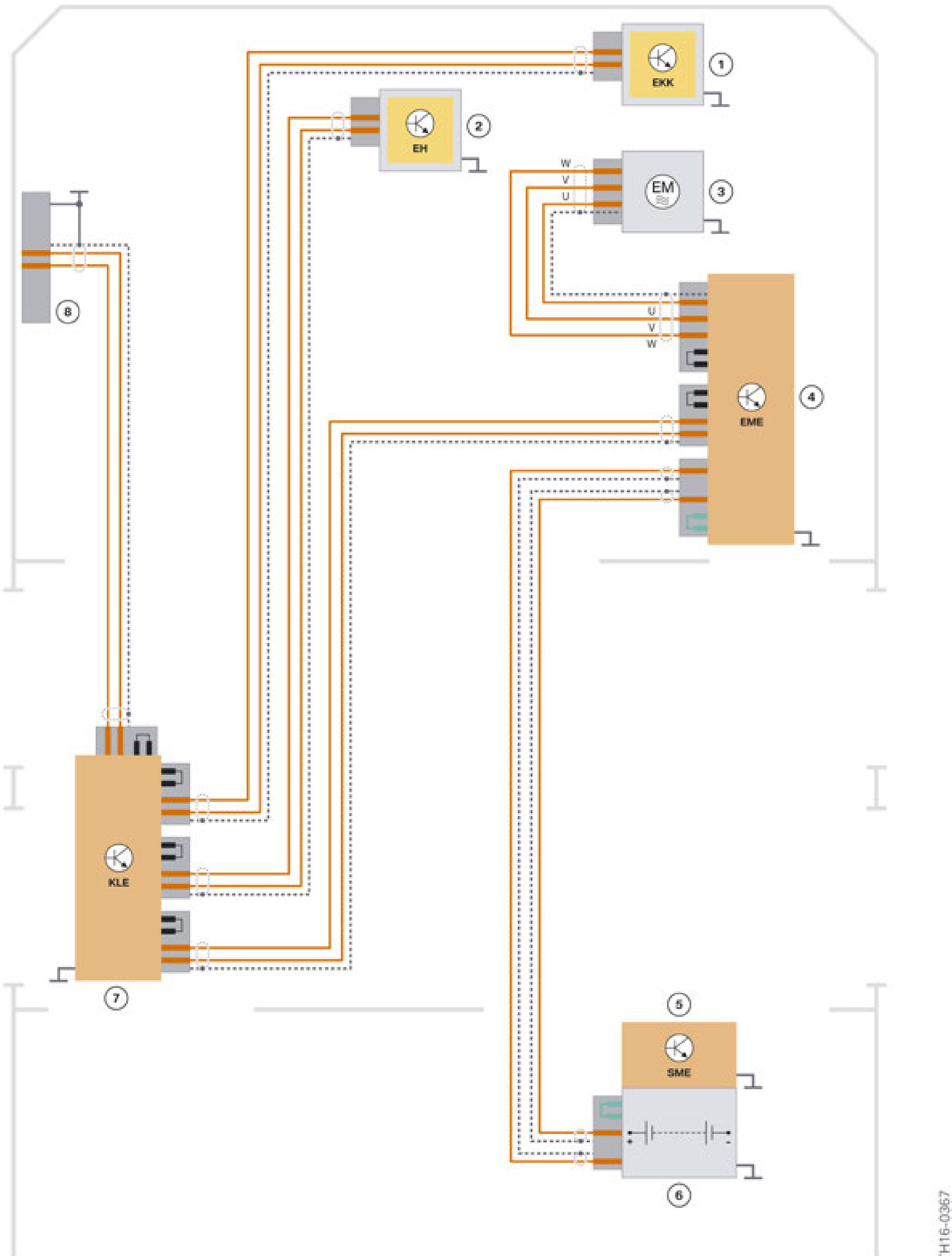
The high-voltage battery unit is located outside the passenger compartment. If the battery cells generate excess pressure due to a massive internal fault, the arising gases cannot be vented outwards via a vent pipe. One venting unit on the high-voltage battery unit housing is sufficient to allow pressure compensation.

Just like in other ActiveHybrid vehicles, the high-voltage service disconnect is not part of the high-voltage battery unit. This component is located in the rear right luggage compartment behind a flap.

G12 PHEV High-voltage Battery Unit

2. High-voltage Battery Unit

2.1.4. System wiring diagram



G12 PHEV system wiring diagram high-voltage battery unit in high-voltage network

Index	Explanation
1	Electric A/C compressor
2	Electrical Heating (EH)
3	Electric motor
4	Electric Motor Electronics (EME)
5	Battery management electronics (SME)
6	Cell modules of the high-voltage battery unit
7	Convenience charging electronics (KLE)
8	Charging socket

2.2. External features

2.2.1. Labels

Three labels are attached to the high-voltage battery unit of the G12 PHEV: 1 type plate and 2 warning stickers. The type plate provides logistical information (e.g. part number) and the key technical data (e.g. nominal voltage). The warning stickers point to the lithium-ion technology on the one hand, and to the high electrical voltage used in the high-voltage battery unit on the other. They raise awareness of the potential dangers involved.

G12 PHEV High-voltage Battery Unit

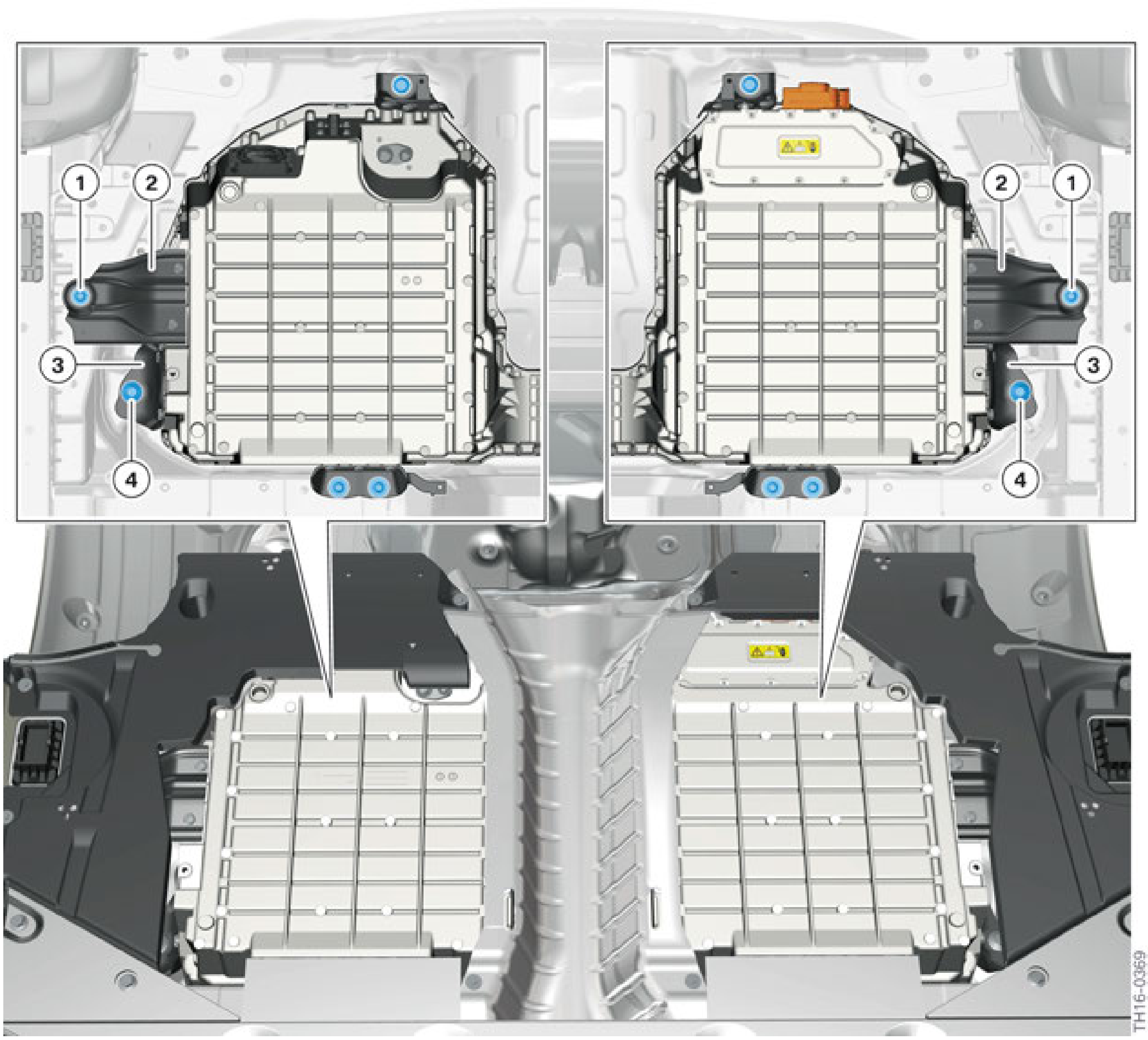
2. High-voltage Battery Unit

G12 PHEV signs on the housing of the high-voltage battery unit

Index	Explanation
1	High-voltage battery unit warning sticker
2	Type plate with technical data
3	Lower housing section of the high-voltage battery unit
4	High-voltage component warning sticker
5	Upper housing section of the high-voltage battery unit

2.2.2. Mechanical interfaces

The high-voltage battery unit is connected to the vehicle body by means of eight holders and 10 mounting bolts. This way the weight and the acceleration forces occurring during driving are supported at the body. All mounting bolts are accessible from the vehicle underbody. However, several trim panels, the exhaust system and the drive shaft need to be removed before the bolts can be loosened.



G12 PHEV mounting of the high-voltage battery unit

Index	Explanation
1	Mounting bolt
2	Brackets (absorb forces in case of an accident)
3	Brackets (absorb forces in normal situations)
4	Mounting bolt and potential compensation bonding screw

2. High-voltage Battery Unit

The two lateral brackets (2) absorb the forces in case of an accident. The other brackets (3) absorb the forces occurring in normal driving situations.

When removing the high-voltage battery unit, all preliminary work specified in the repair instructions (diagnosis, disconnecting from the power supply, drawing off the refrigerant disassembling the trim panels, etc.) must be performed first. Before undoing the mounting bolts, the mobile assembly lifting table with matching supports must be positioned under the high-voltage battery unit.

The electrical connection between the housing of the high-voltage battery unit and the body is established by means of two mounting bolts.

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The low resistance connection between the housing of the high-voltage battery unit and ground is a crucial prerequisite for the fault-free function of the automatic insulation monitoring. This is why it is important to observe the correct tightening torque for all assembly screws.

It is also important to ensure that neither the housing of the high-voltage battery unit, nor the body are painted, corroded or contaminated around the corresponding bore holes. The bare metal must be exposed if necessary before the mounting screws can be secured.

When mounting the potential compensation bonding screws, the exact procedure must be observed:

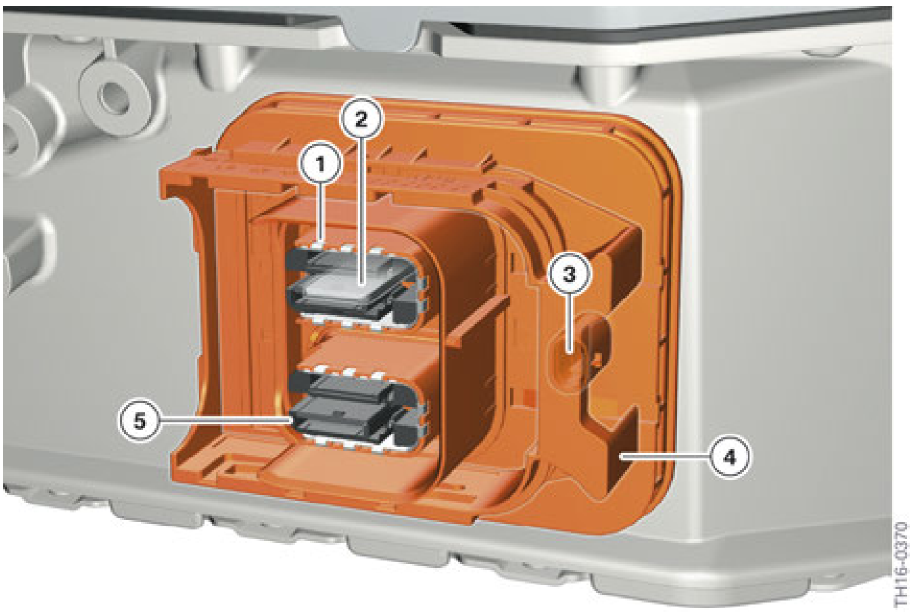
- Clean contact surfaces and have checked by a second person.
- Tighten assembly screws to specified torque.
- Have torque checked by second person.
- Both persons must record this in the vehicle records.

Only self-tapping screws have to be used for assembly procedures at the housing of the high-voltage battery unit. It is permitted to repair the thread on the lower housing section using Kerb Konus threaded inserts.

2.2.3. Electrical interfaces

High-voltage connection

There is a two-pin high-voltage connection at the high-voltage battery unit with which the high-voltage battery unit is connected to the high-voltage electrical system.



G12 PHEV high-voltage connection on the high-voltage battery unit

Index	Explanation
1	Contact for shielding
2	Contact for high-voltage cable
3	Connection bridge in the circuit of the high-voltage interlock loop
4	Mechanical slide
5	Contact protection

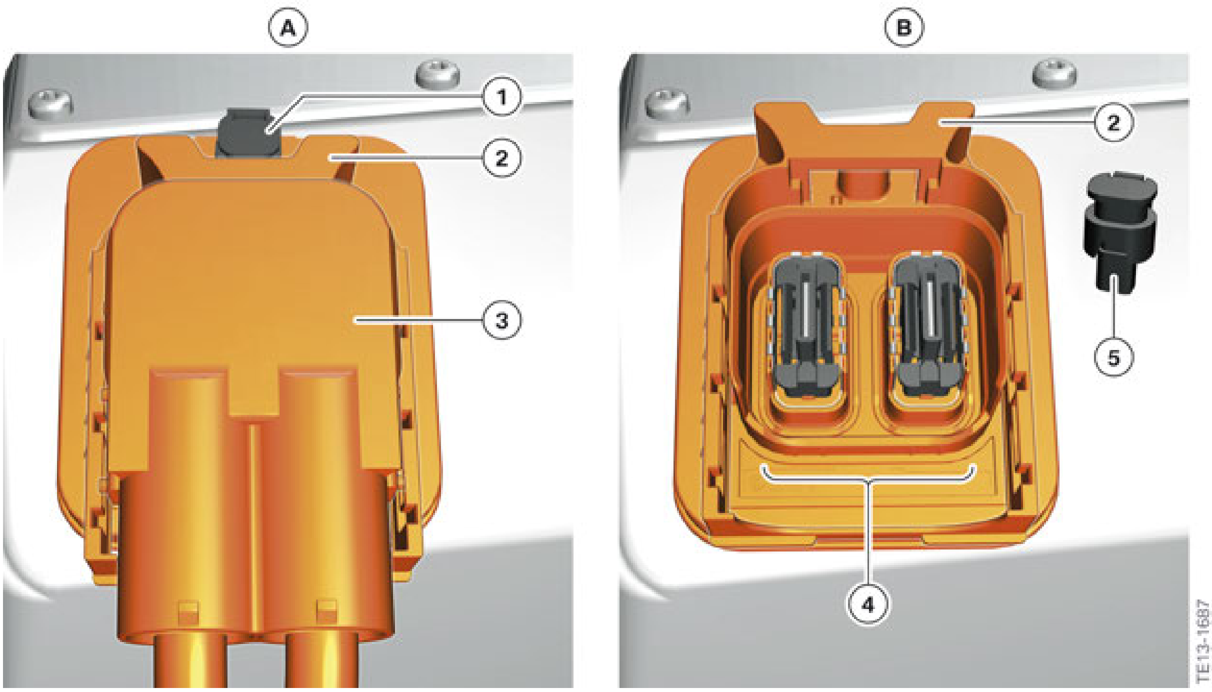
A contact is available for shielding around each of the two electrical contacts of the high-voltage cables. The shielding of the high-voltage cable (shielding for each cable) is continued into the housing of the high-voltage battery unit and thus contributes to the electromagnetic compatibility (EMC).

In addition, the high-voltage connection provides protection against contact with live parts. The actual contacts are coated in plastic so that nobody can touch them directly. Only when the cable is connected is the coating pushed away and contact is made.

The plastic slide serves as the mechanical latch mechanism of the connector. It is also an element of a safety function: If the high-voltage cable is not connected, the slide conceals the connection for the bridge of the high-voltage interlock loop. Only when the high-voltage cable is properly connected and the connector is locked, is this connection accessible and the bridge can be inserted. This guarantees that only when a high-voltage cable is connected is the circuit of the high-voltage interlock loop also closed. This principle applies to all flat high-voltage connections in the G12 PHEV (high-voltage battery unit, Electrical Machine Electronics).

2. High-voltage Battery Unit

As a result, the high-voltage system is active only once all high-voltage connections have been connected to the Electrical Machine Electronics and the convenience charging electronics. This is additional protection against contact with contact surfaces, which may carry voltage.



High-voltage connection (example)

Index	Explanation
A	High-voltage connection with connected high-voltage cable
B	High-voltage connection with disconnected high-voltage cable
1	Bridge for high-voltage interlock loop (connected)
2	Mechanical slide
3	High-voltage connector of the high-voltage cable
4	High-voltage connection
5	Bridge for high-voltage interlock loop (disconnected)

The high-voltage connection can be replaced as a separate component, just like all other components of the high-voltage battery unit. The prerequisites are qualified service technicians and exact compliance with the repair instructions.

2. High-voltage Battery Unit

Interface for the 12 V vehicle electrical system

The high-voltage battery unit of the G12 PHEV features an interface to the 12 V vehicle electrical system with the following connections:

- Voltage supply of the SME control unit with terminal 30 and terminal 31
- Terminal 30C crash signal for voltage supply of the electromechanical switch contactors
- Input and output of the line for the high-voltage interlock loop
- Output (+12 V and ground) for the activation of the combined expansion and shutoff valve
- PT-CAN3

G12 PHEV Signal connector on the high-voltage battery unit

Index	Explanation
1	Interface for the 12 V vehicle electrical system

High-voltage safety connector (Service Disconnect)

The high-voltage service disconnect of the G12 PHEV is not a direct part of the high-voltage battery unit. For this reason, the color of the high-voltage service disconnect was changed from orange to **green** as automotive standard. The high-voltage service disconnect is installed in the rear right luggage compartment as a separate component.

2. High-voltage Battery Unit

G12 PHEV, installation location of the high-voltage service disconnect

The high-voltage service disconnect fulfils two tasks:

- Disconnecting the high-voltage system from the supply
- Providing a safeguard to prevent unintentional restarting

The high-voltage safety connector or the connected bridge is part of the circuit of the high-voltage interlock loop. If the connector of the high-voltage safety connector are pulled apart, the circuit of the high-voltage interlock loop is interrupted.

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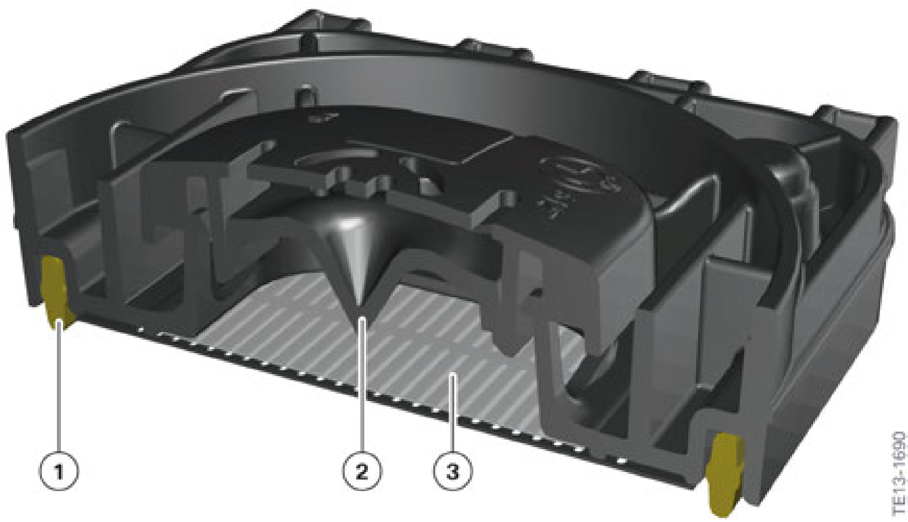
The connection of the high-voltage safety connector cannot be disconnected from each other completely. Both parts are mechanically secured against complete separation. To interrupt the circuit of the high-voltage interlock loop, it is sufficient to separate the two parts far enough so that the padlock can be used to secure against restart.

The precise procedure for disconnecting the power supply is described in the section "G12 PHEV High-voltage Components" .

2.2.4. Venting unit

The venting unit performs two tasks. The first task is to offset large pressure differences between the inside and outside of the high-voltage battery unit. Such pressure differences can only arise in the event of a damaged battery cell. In this case for safety reasons the housing of the cell module with the damaged battery cell is opened to reduce the pressure. The gases are initially located in the housing of the high-voltage battery unit. From there they can be transported outwards via the venting unit. The pressure also increases if the evaporator in the high-voltage battery unit is leaking and refrigerant is escaping.



G12 PHEV cross-section through the venting unit

Index	Explanation
1	Gasket
2	Mandrel
3	Diaphragm

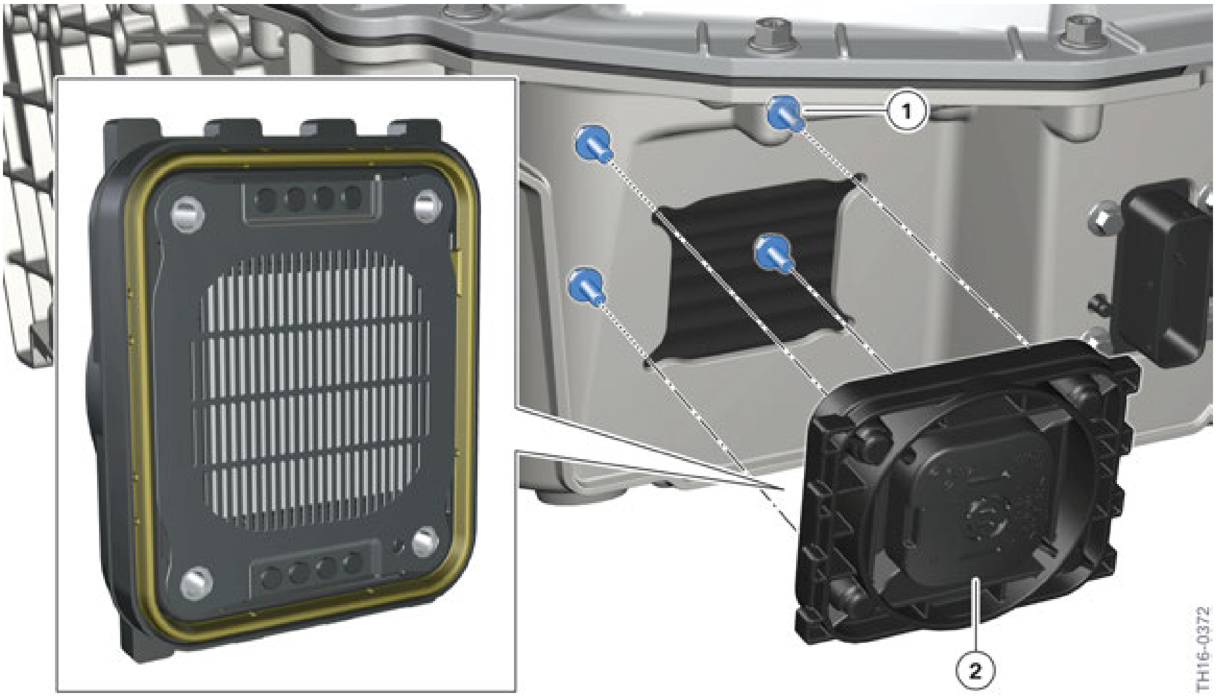
The second task of the venting unit is to remove condensation that might form inside of the high-voltage battery unit. Besides the technical components, there is also air inside the high-voltage battery unit.

If the air or the housing is cooled by a lower ambient temperature or by a refrigerant through the activation of the air conditioning function, some of the water vapor from the air condenses. This means small amounts of liquid water can form in the inside of the high-voltage battery unit.

During the next heating of the air or housing, the water evaporates again and at the same time, the pressure in the housing rises slightly. The venting unit permits pressure compensation by allowing the warm air to escape outwards. The water vapor in the air is also transported outwards.

To fulfil these tasks the venting unit has a permeable diaphragm for gases (and water vapor) and an impermeable diaphragm for fluids. Above the diaphragm is a mandrel, which destroys the diaphragm in the event of severe excess pressure in the high-voltage battery unit. There is a two-part cover to protect against coarse contamination.

The installation location of the venting unit is on the lower housing section.



G12 PHEV mounting, venting unit

Index	Explanation
1	Mounting bolts
2	Venting unit

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The venting unit can be replaced in Service as an entire unit. If the venting unit is mechanically damaged, or the diaphragm is faulty, then the venting unit must be replaced.

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Use the test adapter matching the G12 PHEV venting unit for the final test with the EoS diagnosis system (End of Service).

2.2.5. Interface for the refrigerant circuit

It is incorporated in the refrigerant circuit of the heating and air-conditioning system for cooling the high-voltage battery. In order to be able to perform condition-based cooling, there is an electrically combined expansion and shutoff valve at the high-voltage battery unit.

The combined expansion and shutoff valve is hard-wired to the SME control unit and is activated directly by this control unit. The valve is closed in a currentless state, i.e. no refrigerant flows into the high-voltage battery unit. The valve only knows two positions, "closed" and "open". The amount of flowing refrigerant is adjusted by the thermal portion of the expansion valve.

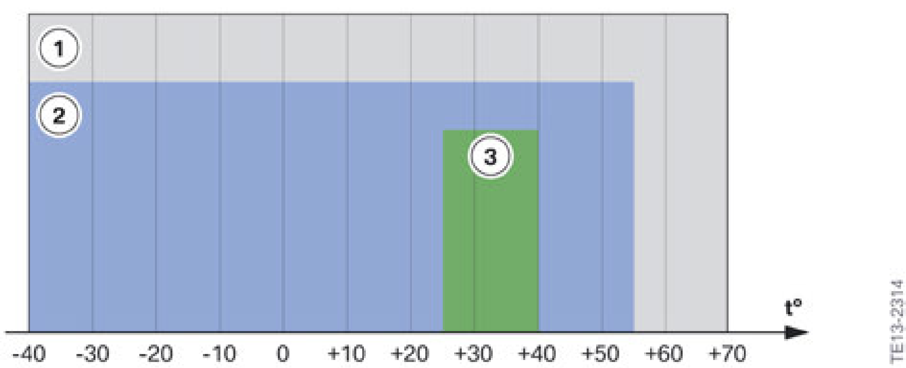
2.3. Cooling system

2.3.1. Overview

To maximize the service life of the high-voltage battery and obtain the greatest possible power, it is operated in a defined temperature range. The high-voltage battery unit is fundamentally operational in the range of -40 °C to +55 °C (-40 °F to + 131 °F) (actual cell temperature). In terms of temperature behavior the high-voltage battery unit is a slow-action system, i.e. it takes several hours until the cells assume the ambient temperature. Therefore having the battery spend a short period of time in an extremely hot or cold environment does not mean that the cells will already have assumed this temperature.

The optimal range of the temperature of the cell with regard to service life and performance is limited. It is between +25 °C (+77 °F) and +40 °C (+104 °F). If the cell temperature is continuously significantly above this range with simultaneously high performance, this has a negative effect on the service life of the battery cells. To counteract this effect, as well as ensure maximum performance at all ambient temperatures, the high-voltage battery unit of the G12 PHEV has an automatic cooling function.

There is no heating function for the high-voltage battery unit in the G12 PHEV.

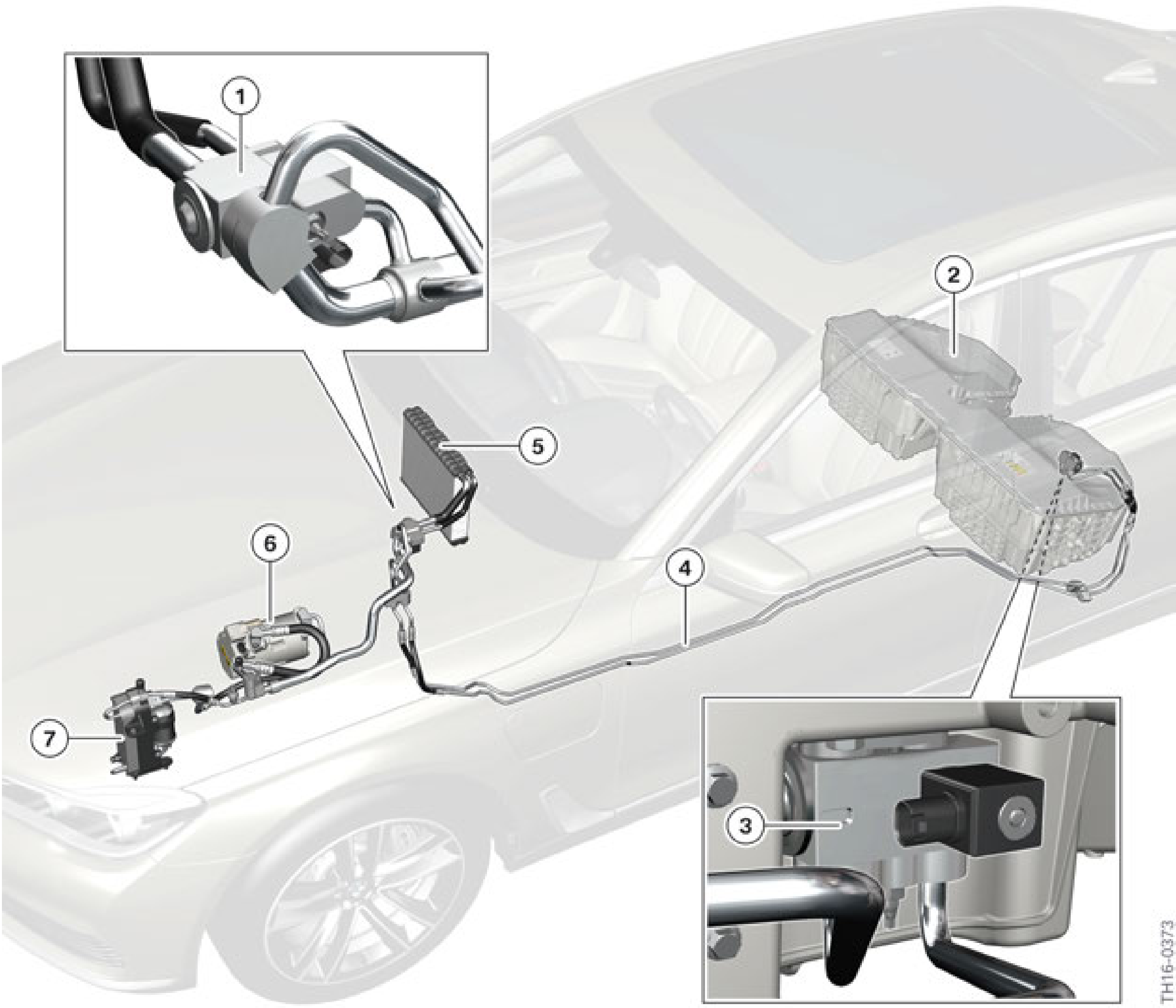


G12 PHEV high-voltage battery unit operating temperature range

Index	Explanation
1	General temperature range (storage area)
2	Operating range of the high-voltage battery unit
3	Optimal operating range of the high-voltage battery unit

The G12 PHEV is equipped with a cooling system for the high-voltage battery as standard. For this purpose it is integrated in the refrigerant circuit of the heating and air-conditioning system.

2. High-voltage Battery Unit



G12 PHEV Overall refrigerant system

Index	Explanation
1	Combined expansion and shutoff valve
2	High-voltage battery unit
3	Combined expansion and shutoff valve
4	Refrigerant lines to high-voltage battery unit
5	Evaporator, passenger compartment
6	Electric A/C compressor
7	Coolant-cooled air conditioning condenser (Coolant/refrigerant heat exchanger)

Cooling system of the high-voltage battery unit

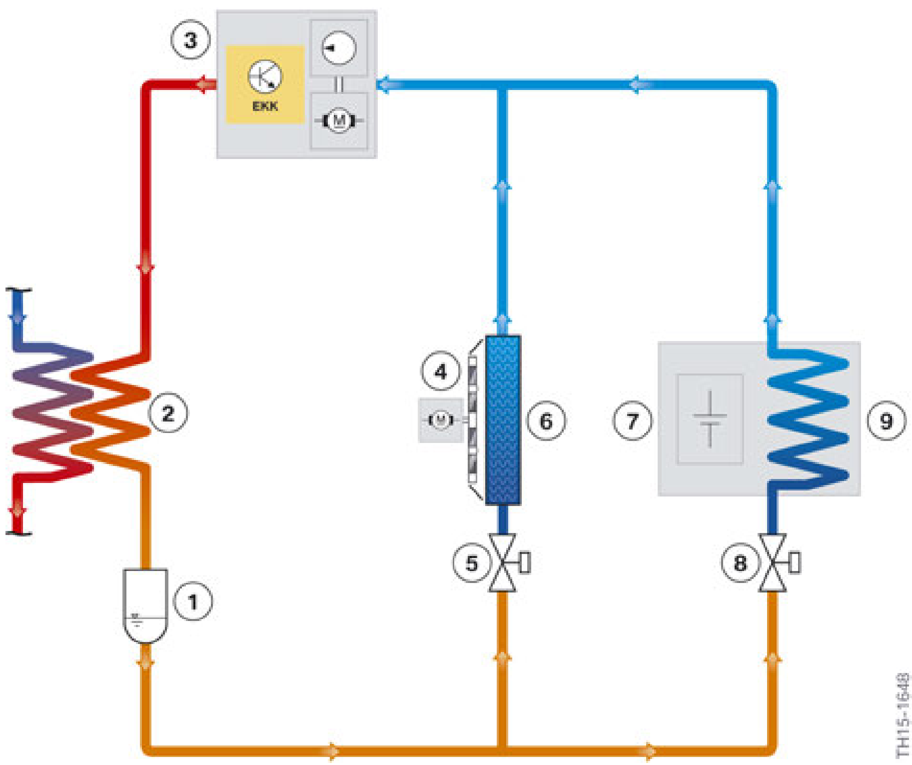
The high-voltage battery unit in the G12 PHEV is cooled directly using R1234yf refrigerant.

G12 PHEV High-voltage Battery Unit

2. High-voltage Battery Unit

The refrigerant circuit of the air conditioning consists of two parallel branches. One for cooling the passenger compartment and one for cooling the high-voltage battery unit.

For each branch there is a combined expansion and shutoff valve in order to be able to control the cooling functions independent of each other. The battery management electronics can activate and open the combined expansion and shutoff valve at the high-voltage battery unit by applying voltage. In this way refrigerant can flow to the high-voltage battery, where it expands, evaporates and cools the surrounding area. The cooling of the passenger compartment is also effected in a condition-based manner. The combined expansion and shutoff valve upstream from the evaporator can also be activated electrically and by the EME.



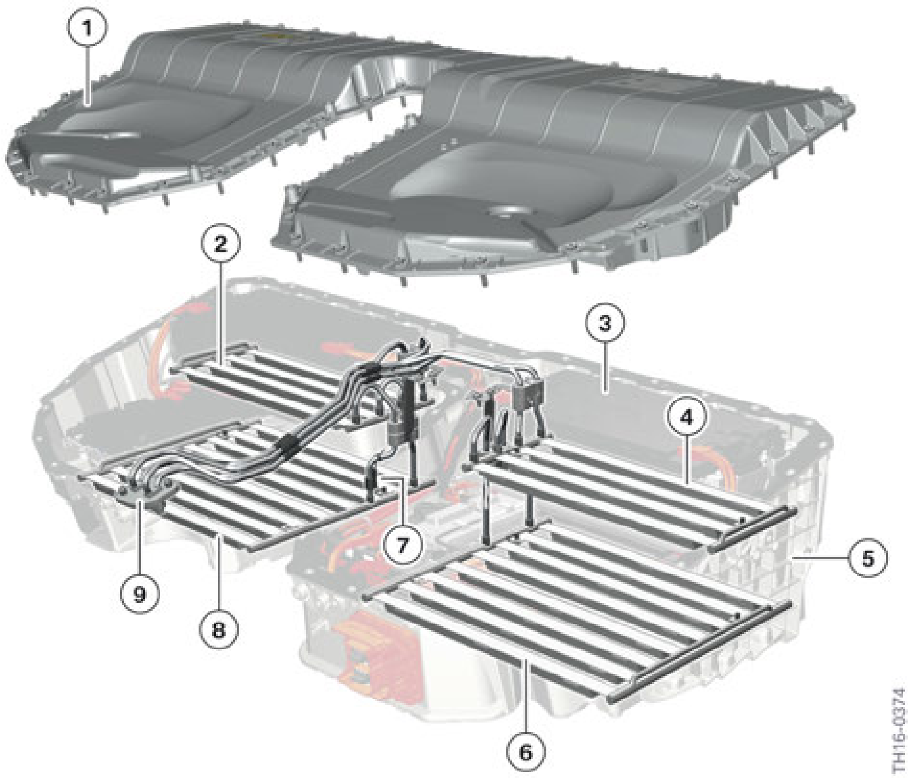
G12 PHEV refrigerant circuit with high-voltage battery unit (simplified illustration)

Index	Explanation
1	Receiver dryer
2	Coolant-cooled air conditioning condenser (Coolant/refrigerant heat exchanger)
3	Electric A/C compressor
4	Blower for passenger compartment
5	Combined expansion and shutoff valve, passenger compartment
6	Evaporator, passenger compartment
7	High-voltage battery unit
8	Combined expansion and shutoff valve, high-voltage battery unit
9	High-voltage battery unit evaporator

2. High-voltage Battery Unit

The refrigerant evaporates due to injecting liquid refrigerant into the heat exchanger. The evaporating refrigerant draws the heat from the ambient air cooling it. Then the EKK compresses the refrigerant at a higher pressure level. The air conditioning condenser then transfers heat energy to the coolant, and the refrigerant again returns to a liquid state.

In the G12 PHEV, the cell modules are arranged on two levels due to the installation location of the high-voltage battery unit. A four-part evaporator is used to ensure sufficient cooling of the cell modules, for a given arrangement. It consists of flat aluminum pipes and is connected to the internal refrigerant lines.



G12 PHEV components for cooling in the high-voltage battery unit

Index	Explanation
1	Upper housing section
2	Heat exchanger (top left viewed in direction of travel)
3	Cell module
4	Evaporator (top right)
5	Lower housing half
6	Evaporator (bottom right)
7	Temperature sensor for refrigerant line
8	Evaporator (bottom left)
9	Connecting flange for combined expansion and shutoff valve

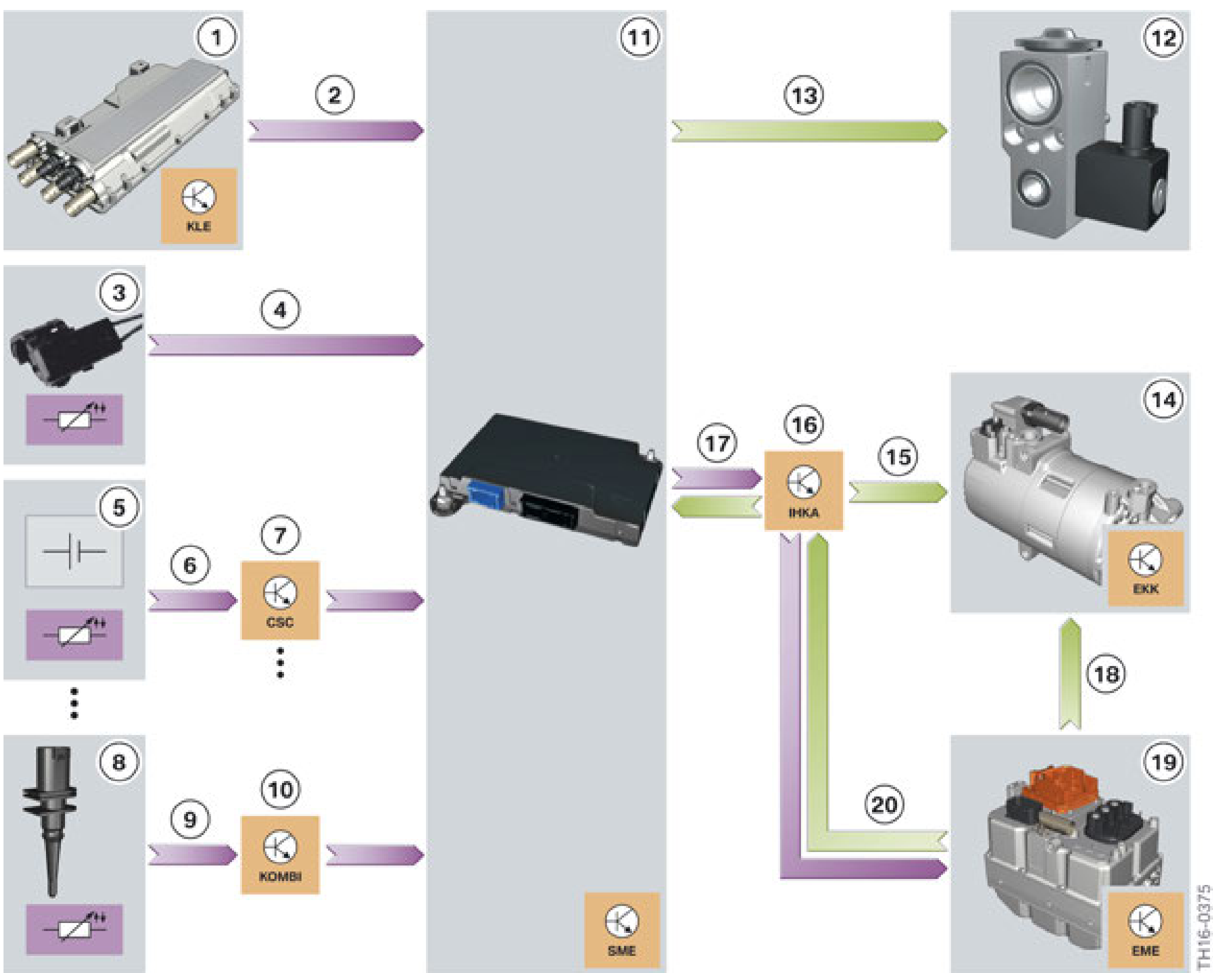
2.3.2. Function

The cooling system has two operating conditions:

- Cooling OFF
- Cooling ON

The operating conditions are generally dependent on the cell temperatures, the ambient temperature and the power used by or supplied to the high-voltage battery. The SME control unit decides which operating condition is necessary depending on the input variables. When cooling is needed, the combined expansion and shutoff valve is activated and opened by the battery management electronics. Otherwise it remains closed.

The following graphic shows the input variables, the role of the SME control unit and which actuators are used for the control.



Input/output refrigerant system of the high-voltage battery unit G12 PHEV

Index	Explanation
1	Convenience charging electronics (KLE)
2	Information that the high-voltage battery unit is being charged externally
3	Refrigerant temperature sensor at the refrigerant line supply
4	Signal for refrigerant temperature
5	Temperature sensors at the cells of the high-voltage battery
6	Signal for cell module temperature
7	Cell Supervision Circuit (CSC)
8	Outside temperature sensor
9	Signal for ambient temperature
10	Instrument panel (KOMBI)
11	SME control unit (in the high-voltage battery unit)
12	Combined expansion and shutoff valve
13	Signal for activating the combined expansion and shutoff valve
14	Electric A/C compressor
15	Signal via LIN bus for activating the EKK
16	Integrated automatic heating / air conditioning (IHKA)
17	Cooling requirement
18	High-voltage power supply
19	Electric Motor Electronics (EME)
20	High-voltage power requirement

Cooling OFF operating condition

The "Cooling OFF" operating condition is adopted when the cell temperatures are in or below an optimal range. This is generally the case if the vehicle is moved at moderate ambient temperatures and at low electrical power. The "Cooling OFF" operating condition is particularly efficient because no additional energy is required for cooling the high-voltage battery.

The components involved work as follows:

- The EKK is not in operation or runs at reduced power if the passenger compartment has to be cooled.
- The combined expansion and shutoff valve at the high-voltage battery unit is closed.

Cooling ON operating condition

If the battery cells heat up to temperatures of approx. 30 °C (86 °F), the cooling of the high-voltage battery has already begun. The SME control unit sends a cooling requirement to the IHKA control unit with two priorities. The IHKA decides whether the passenger compartment or the high-voltage battery unit is cooled or both are cooled. The cooling requirement may be rejected by the IHKA for a cooling requirement by the SME with low priority and for a high cooling requirement in the passenger compartment. However, in case of a high priority cooling request from the battery management electronics, the high-voltage battery unit is always cooled.

The integrated automatic heating/air conditioning system requests electrical power for cooling the EKK from the high voltage power management in the electrical machine electronics.

The components act as follows in "Cooling" operating condition:

- SME control unit requests cooling requirement.
- Following release by the IHKA, the SME control unit activates the combined expansion and shutoff valve at the high-voltage battery unit. This valve opens and refrigerant flows to the high-voltage battery unit.
- The EKK is operational.

As a result of the pressure drop after the expansion valve, the refrigerant in the lines and coolant ducts of the high-voltage battery unit evaporates. In the process the refrigerant absorbs heat energy from the cell modules or battery cells and cools them. The evaporated refrigerant then leaves the high-voltage battery unit, is compressed by the EKK and returned to a liquid state in the capacitor. Although energy is required from the high-voltage electrical system for this procedure, it is of utmost importance. Only this way can a long service life and a high degree of efficiency of the battery cells be guaranteed.

If the temperature of the battery cells is significantly below the optimal operating temperature of 20 °C (68 °F), their performance would be temporarily restricted and the efficiency of the energy conversion would not be optimal. This is an unavoidable chemical effect of lithium-ion batteries.

If the G12 PHEV is parked for an extended period (e.g. a few days) at a very low ambient temperature, the battery cells also take on this low ambient temperature. In this situation it is possible the full electrical driving power is not available at the start of the trip. This remains unnoticed by the customer as the combustion engine assumes the drive of the vehicle.

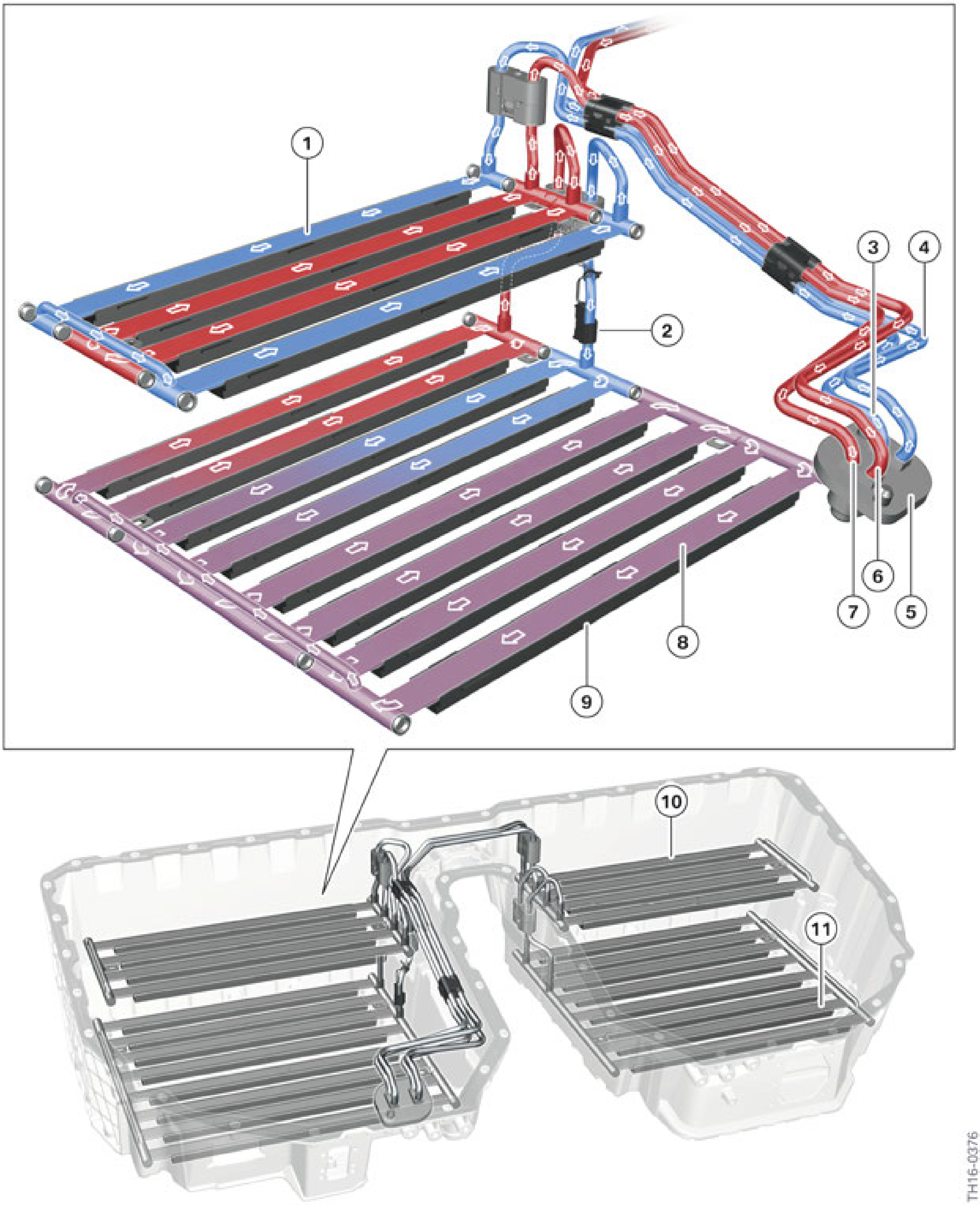
2.3.3. System components

Evaporator

In the inside of the high-voltage battery units the refrigerant flows through lines and coolant ducts made from aluminum. The refrigerant flowing in via the inlet piping is distributed downstream of the connection to the high-voltage battery unit to the left and right evaporators. The refrigerant flowing through the feed line is distributed in the evaporators and absorbs the heat of the cell modules en route through the coolant ducts.

Finally, the right and left evaporators are merged into a shared return line. This common return line returns the evaporated refrigerant to the connection of the high-voltage battery unit.

A temperature sensor is installed on the feed line of the lower left evaporator; its signal is used for controlling and monitoring the cooling function. It is read directly by the SME control unit.



G12 PHEV components for cooling in the high-voltage battery unit

2. High-voltage Battery Unit

Index	Explanation
1	Evaporator (top left, viewed in direction of travel)
2	Refrigerant temperature sensor
3	Feed line for left evaporator pressure stage (viewed in direction of travel)
4	Feed line for right evaporator pressure stage
5	Connecting flange for combined expansion and shutoff valve
6	Return line intake side for right evaporator
7	Return line intake side for left evaporator
8	Evaporator (bottom left)
9	Spring strip
10	Evaporator (top right)
11	Evaporator (top left)

So that the coolant ducts can fulfil their task of discharging heat energy from the cell modules, the whole area of the coolant ducts must be pressed onto the cell modules at an evenly distributed force. This contact pressure is generated by spring rods which are embedded in the coolant ducts.

The spring strips of the bottom evaporators are supported by the lower housing section of the high-voltage battery unit, and press the coolant ducts against the cell modules. The spring strips of the top evaporators are supported by the module intermediate flooring panel.

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The refrigerant lines, coolant ducts and spring rails together form one unit, which can be replaced separately in the case of a repair. To simplify matters, this unit is also referred to as an evaporator, but must not to be confused with the heat exchanger/radiator at the front of a non PHEV vehicle.

The four evaporators are components with a relatively thin wall thickness. They have excellent heat conducting properties, but as a result their mechanical stability is low. This is not a disadvantage when it is installed because the mechanical stability is maintained by the high-voltage battery unit housing. However, extreme caution must be taken when handling the evaporator during Service.

⚠

When replacing the evaporator, precisely follow the repair instructions to the word, and use extreme caution.

Refrigerant temperature sensor

The temperature of the refrigerant is not measured directly. Instead, the temperature sensor is fitted at a segment of the refrigerant line in the high-voltage battery unit. The installation position is shown in the graphic in the chapter entitled "Evaporator".

Using the temperature of the refrigerant line, the temperature of the flowing refrigerant can be calculated. The refrigerant temperature sensor is hard-wired to the SME control unit where the signal is evaluated.

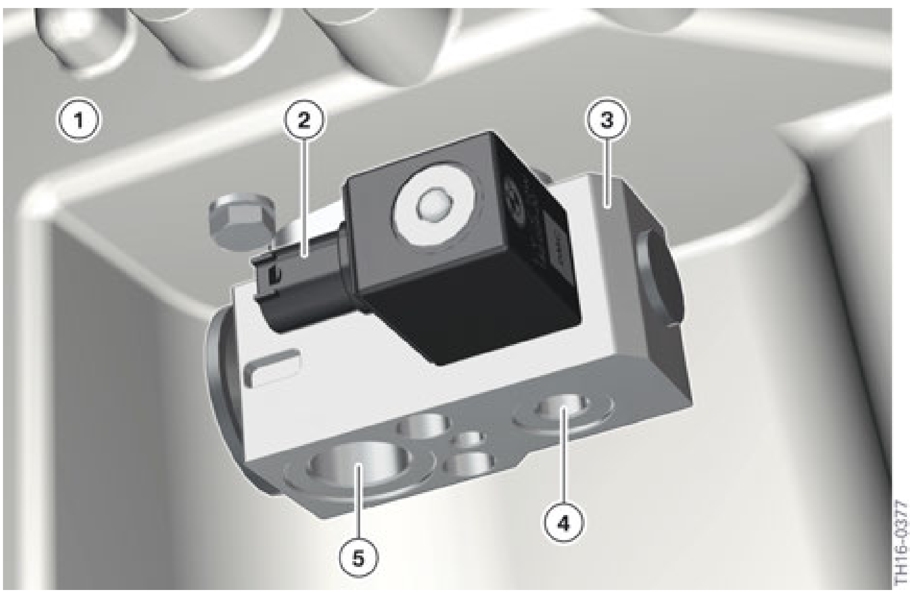
The sensor is a Negative Temperature Coefficient (NTC), whose resistance value is reduced at increasing temperature.

The refrigerant temperature sensor can be replaced individually in the event of a fault.

Combined expansion and shutoff valve

The combined expansion and shutoff valve causes the in-flowing liquid refrigerant to expand, reducing its pressure and temperature. Part of the refrigerant evaporates in this process. Due to the reduction in temperature, the refrigerant can absorb ambient heat and cool the cell modules.

Additionally, the SME can shut off the refrigerant circuit via the shutoff valve located on the expansion valve so no more refrigerant flows into the evaporator.



G12 PHEV combined expansion and shutoff valve

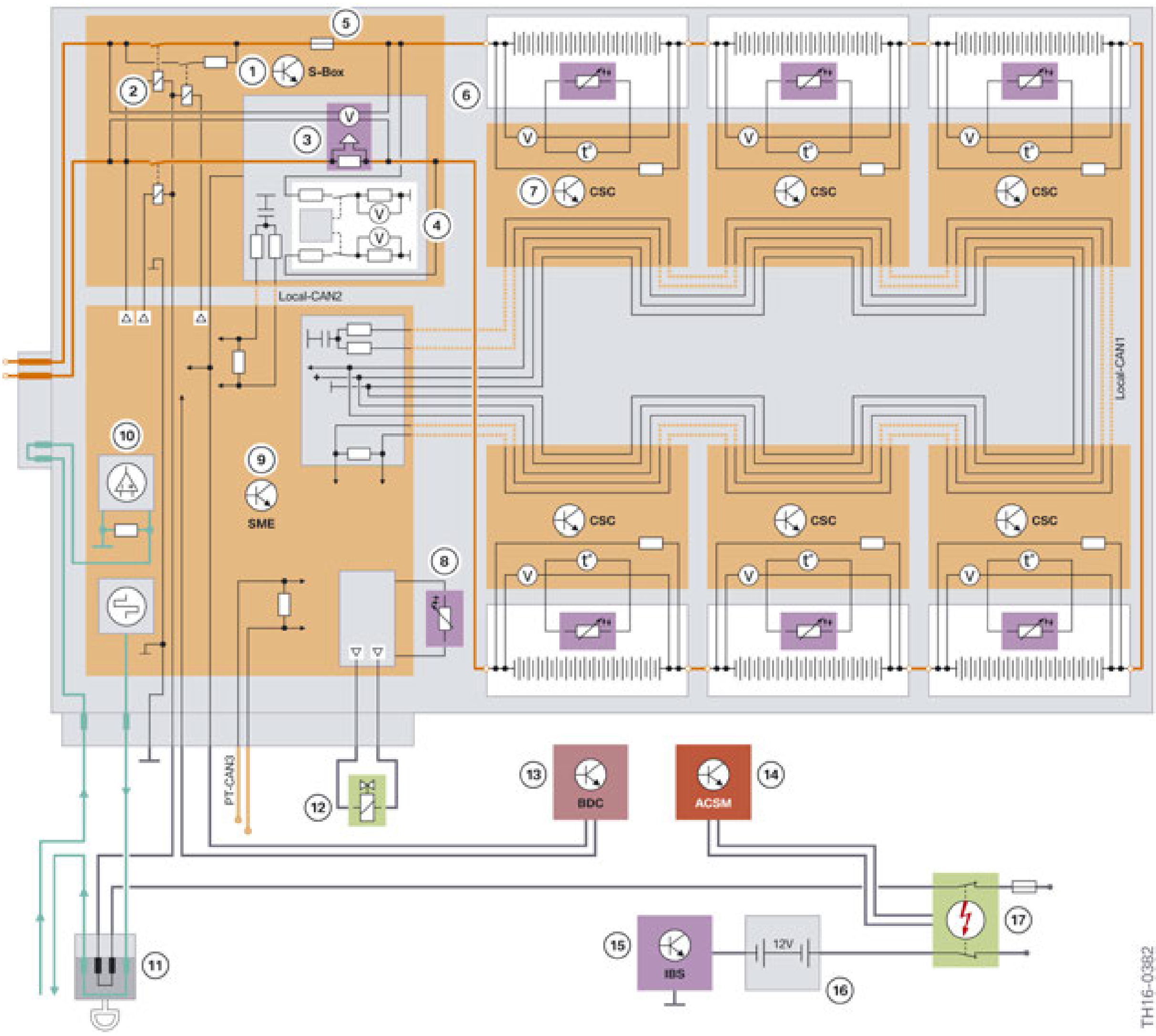
Index	Explanation
1	Lower housing section of the high-voltage battery unit
2	Electrical connection for combined expansion and shutoff valve
3	Combined expansion and shutoff valve
4	Connection for refrigerant pressure line
5	Connection for refrigerant intake pipe

The combined expansion and shutoff valve is activated by the battery management electronics control unit using direct wiring.

The electrical activation has two conditions: A voltage of 0 V means that the valve is closed. A voltage of 12 V opens the valve. Similar to conventional expansion valves in heating and air-conditioning systems, this expansion and shutoff valve also automatically adjusts its degree of opening thermally, i.e. depending on the refrigerant temperature.

2.4. Inner structure

2.4.1. Electrical and electronic components



G12 PHEV, system wiring diagram for high-voltage battery unit

Index	Explanation
1	Safety box
2	Switch contactors
3	Current and voltage sensor
4	Isolation monitoring
5	Main current fuse (350 A)
6	Cell module
7	Cell Supervision Circuit (CSC)

2. High-voltage Battery Unit

Index	Explanation
8	Temperature sensor for refrigerant line
9	Battery management electronics (SME)
10	Control of the circuit of the high-voltage interlock loop
11	High-voltage safety connector ("Service Disconnect")
12	Combined expansion and shutoff valve of the refrigerant line
13	Body Domain Controller (BDC)
14	ACSM with control lines for activating the safety battery terminal
15	Intelligent Battery Sensor (IBS)
16	12 V battery
17	Safety battery terminal (SBK)

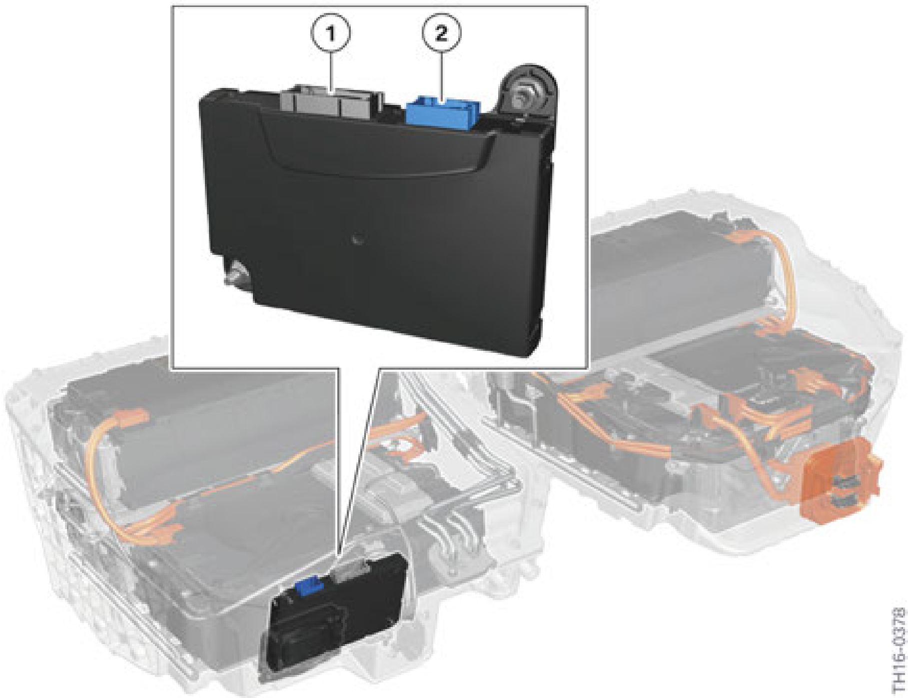
It can be seen from the wiring diagram shown above that the high-voltage battery unit contains the following electrical/electronic components, in addition to the actual battery cells which are grouped in six cell modules:

- Control unit for battery management electronics (SME)
- Six Cell Supervision Circuits (CSC)
- Safety box with switch contactors, sensors, overcurrent fuse and isolation monitoring

In addition to the electrical components, the high-voltage battery unit is also made up of refrigerant lines and coolant ducts, as well as mechanical retaining elements for the cell modules. These internal components, as well as the high-voltage connection and the high-voltage safety connector, are described in the following chapters.

2. High-voltage Battery Unit

Battery management electronics (SME)



G12 PHEV, installation location of the battery management electronics

Index	Explanation
1	Connection, communication wiring harness
2	Connection, Cell Supervision Circuit wiring harness

High demands are placed on the service life of the high-voltage battery (service life of the vehicle). It is not possible to operate the high-voltage battery according to personal preferences if these requirements must be satisfied. Instead, the high-voltage battery is operated in a precisely defined range so as to maximize its service life and performance. This includes the following marginal conditions:

- Operate battery cells in the optimal temperature range (by cooling and if required restricting the current level).
- Adjusting the State of Charge of the individual cells where necessary to one another.
- Use storable energy of the battery in a certain range.

To comply with these marginal conditions, the battery management electronics (SME), is installed in the high-voltage battery unit of the G12 PHEV.

2. High-voltage Battery Unit

The SME control unit must fulfil the following tasks:

- Controlling the starting and shutting down of the high-voltage system at the request of the Electrical Machine Electronics (EME).
- Evaluation of the measurement signals for voltage and temperature of all battery cells, as well as the current level in the high-voltage circuit.
- Control of the cooling system for the high-voltage battery units.
- Determining the State of Charge (SoC) and the State of Health (SoH) of the high-voltage battery.
- Determining the available power of the high-voltage battery and where necessary requesting a limitation from the electric motor electronics.
- Safety functions (e.g. voltage and temperature monitoring, high-voltage interlock loop).
- Identification of faults, storing fault code entries and communication of faults to the Electrical Machine Electronics.

The SME control unit can generally be accessed and programmed by the diagnosis system. For troubleshooting it is important to know that not only control unit faults can be entered in the fault memory of the SME control unit, but also links to faults of other components in the high-voltage battery unit. The fault code entries can be divided into different categories which are dependent on their severity and the available functionality:

- Immediate shutdown of the high-voltage system:
  If the safety of the high-voltage system is affected by the fault or there is a risk that the high-voltage battery may suffer damage as a result of the fault, the high-voltage system is shut down immediately and the contacts of the electromechanical switch contactors are opened.
- Restricted performance:
  If the high-voltage battery is no longer able to supply full power or full energy, the drive power and the range are restricted to protect the components. In this case, drivers can always continue with considerably reduced drive power for a short period of time.
- Fault without direct effect for the customer:
  If, for example, communication between SME or CSC control units is disrupted for a short time, this does not result in a functional restriction or put the safety of the high-voltage system at risk. This is why only one fault code entry is generated which must be analyzed by the BMW Service technician using the diagnosis system. A Check Control message does not appear. The functionality for the customer is not restricted.

The SME control unit is not accessible from outside the high-voltage battery unit. To replace a SME control unit in the event of a fault, the high-voltage battery unit must be removed and opened.

Only qualified technicians that have a certification for Gen 3.0 are permitted to open the high-voltage battery unit. The repair instructions must also be followed exactly and it is particularly important to complete the prescribed checks prior to opening the unit.

2. High-voltage Battery Unit

The electrical interfaces of the SME control unit are:

- 12 V supply for SME control unit (terminal 30 from the front left power distribution box and terminal 31)
- 12 V supply for switch contactors (terminal 30C crash signal)
- PT-CAN3
- Local CAN1 and 2
- Input and output for high-voltage interlock loop
- Line for activating the combined expansion and shutoff valve in the cooling circuit
- Refrigerant temperature sensor

The switch contactors in the high-voltage battery unit are supplied with voltage via a special 12 V line. This line is called terminal 30 crash message, **terminal 30C for short.** The **C** in the terminal designation indicates that this 12 V voltage is switched off in the event of an accident (crash). This line is a (second) output of the safety battery terminal. This means that when the safety battery terminal is activated this supply lead is also interrupted.

This line also runs through the high-voltage safety connector so that the supply to the switch contactors is interrupted also when the high-voltage system is disconnected from the power supply. In both these cases the two switch contactors in the high-voltage battery unit are opened automatically.

The Local CAN1 connects the SME control unit to the Cell Supervision Circuits (CSC) (see next chapter). The Local CAN2 is used for communication between SME control unit and safety box. For example, the information on the measured current level is transmitted via this data bus.

Cell modules

The high-voltage battery unit is made up of six cell modules wired in series. Only one Cell Supervision Circuit is assigned to each cell module. The cell module itself is made up of 16 cells connected in series. Each cell has a nominal voltage of 3.66 V and a nominal capacity of 26 Ah. The cell module order is defined and at the front on the bottom on the left side (viewed in direction of travel).

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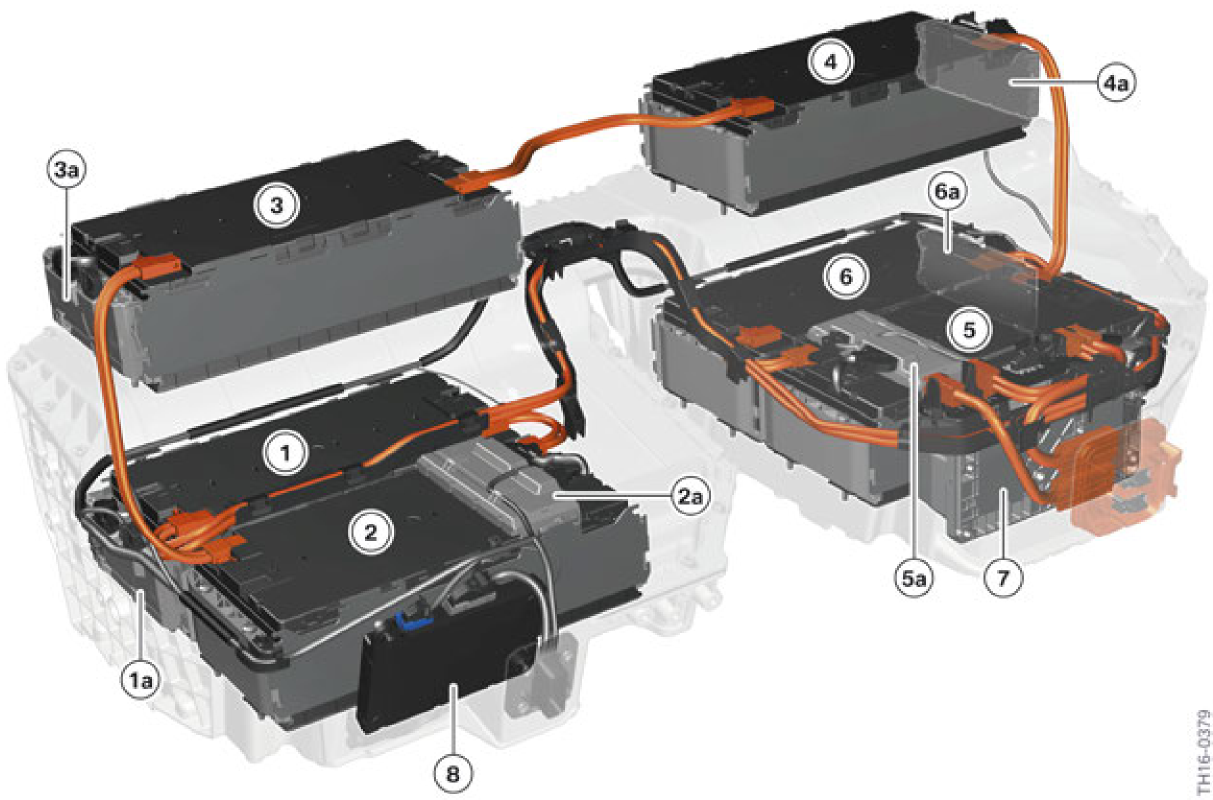
Due to the electrical connection of the cell modules and the arrangement of the Cell Supervision Circuits, there are two different cell module types with respect to polarity. **This must be taken into account for replacement part orders!**

2. High-voltage Battery Unit



G12 PHEV cell module arrangement

Index	Explanation
1	Cell module 1
1a	Cell Supervision Circuit 1a
2	Cell module 2
2a	Cell Supervision Circuit 2a
3	Cell module 3
3a	Cell Supervision Circuit 3a
4	Cell module 4
4a	Cell Supervision Circuit 4a
5	Cell module 5
5a	Cell Supervision Circuit 5a
6	Cell module 6
6a	Cell Supervision Circuit 6a
7	Safety box
8	Battery management electronics (SME)

⚠

When exchanging the cell modules, the assignments to the respective Cell Supervision Circuits must be observed as a mandatory condition, as the assignment is stored in the diagnosis system and referenced for future evaluations.

Cell monitoring

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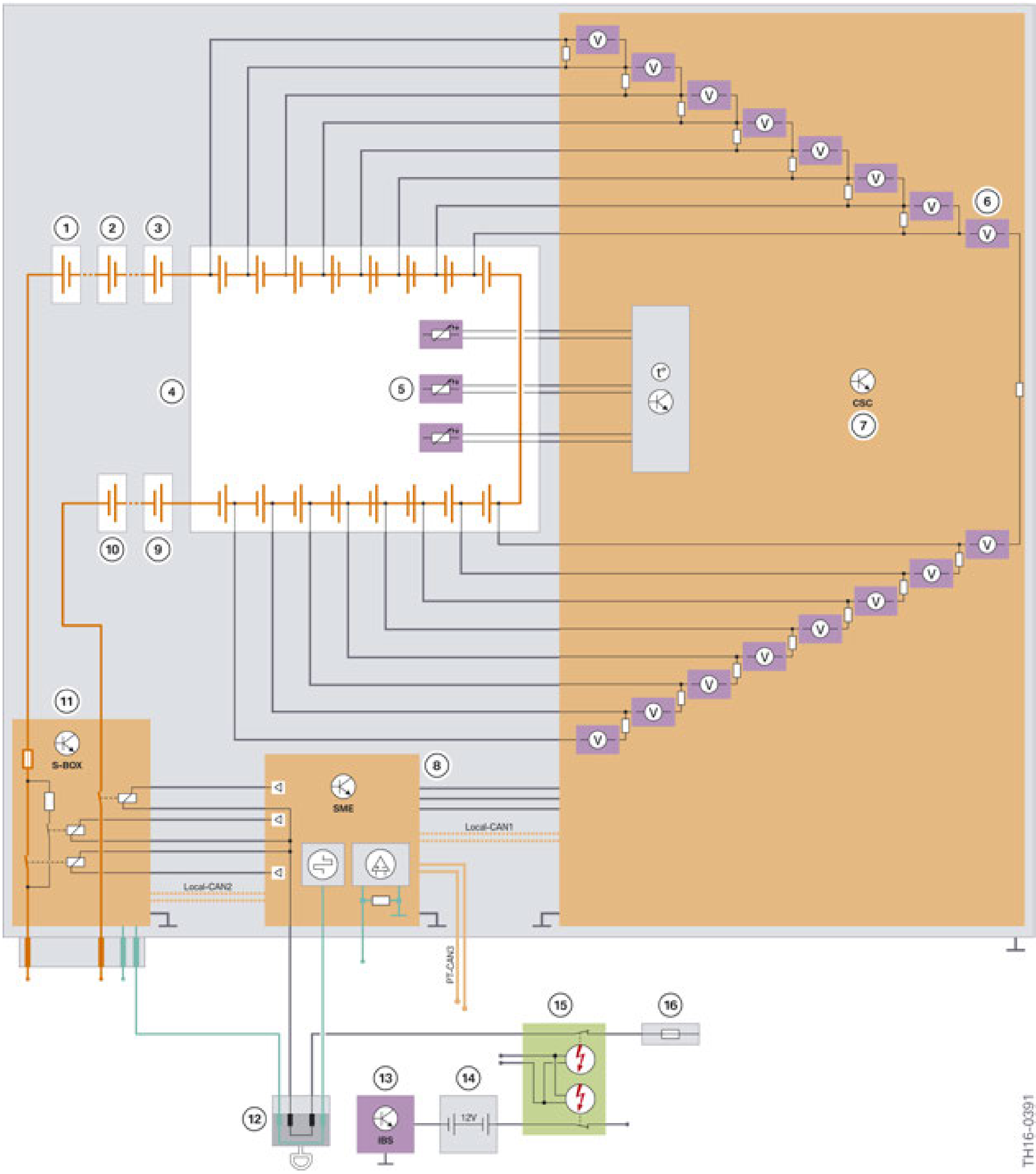
The Cell Supervision Circuits are located inside each high-voltage battery unit. For this reason, only qualified technicians are authorized for performing repairs.

Certain marginal conditions must be observed for fault-free operation of the lithium-ion cells used in the G12 PHEV: The cell voltage and the cell temperature cannot exceed or drop below certain values as otherwise the battery cells may suffer long-term damage. For this reason the high-voltage battery unit features six Cell Supervision Circuits (CSCs).

For each cell module, the high-voltage battery unit of the G12 PHEV has one Cell Supervision Circuit. It monitors 16 cells.



G12 PHEV Cell Supervision Circuit

Index	Explanation
1	Cell module 1
2	Cell module 2
3	Cell module 3
4	Cell module 4

Index	Explanation
5	Temperature sensors at the cell module
6	Voltage measurement of the battery cells
7	Cell Supervision Circuit on module 4
8	Battery management electronics (SME)
9	Cell module 5
10	Cell module 6
11	Safety box
12	High-voltage safety connector ("Service Disconnect")
13	Intelligent Battery Sensor (IBS)
14	12 V battery
15	Safety battery terminal (SBK)
16	Power distribution box, front left

The Cell Supervision Circuits perform the following functions:

- Measurement and monitoring of the voltage of each individual battery cell.
- Measurement and monitoring of the temperature at several points of the cell module.
- Communication of the measured variables to the battery management electronics control unit.
- Implementation of the process for adjusting the cell voltage of the battery cells.

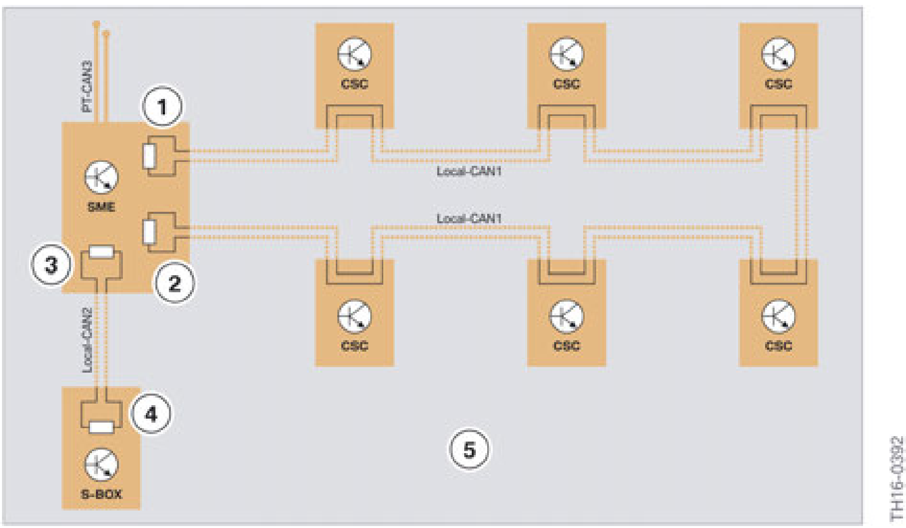
The measurement of the cell voltage is effected at a very high sampling rate (one measurement every 20 milliseconds). The end of the charging procedure, as well as the discharging procedure, can be identified using the voltage measurement. The temperature sensors are arranged at the cell modules so that the temperature of the individual battery cells can be determined from their measured values. Using the cell temperature an overload or electrical fault can be identified. In such a case the current level must be reduced immediately or the high-voltage system shut down completely in order to avoid progressive damage to the battery cells. In addition, the measured temperature is used to control the cooling system in order to constantly operate the battery cells in the temperature range which is optimal for their performance and service life. Because the cell temperature is such an important variable, three NTC temperature sensors are installed for each cell module, of which one is redundant.

The Cell Supervision Circuits communicate the values measured by them via the Local CAN1. This local CAN1 connects all Cell Supervision Circuits to each other, as well as to the SME control unit. In the SME control unit the evaluation of the measured values takes place and a response is introduced if required (e.g. control of the cooling system).

Local CAN1 and 2 operate at a data transfer rate of 500 kBit/s. The bus lines are twisted, as is usual for CAN buses with this data transfer rate. Both Local CANs are also terminated at their ends. The necessary terminating resistors of the Local-CAN 1, each with 120 Ω of the Local CAN1 are located in the SME control unit.

The terminating resistors of the Local-CAN 2, each with 120 Ω of the Local CAN2 are located:

- In the battery management electronics control unit.
- In the safety box control unit.



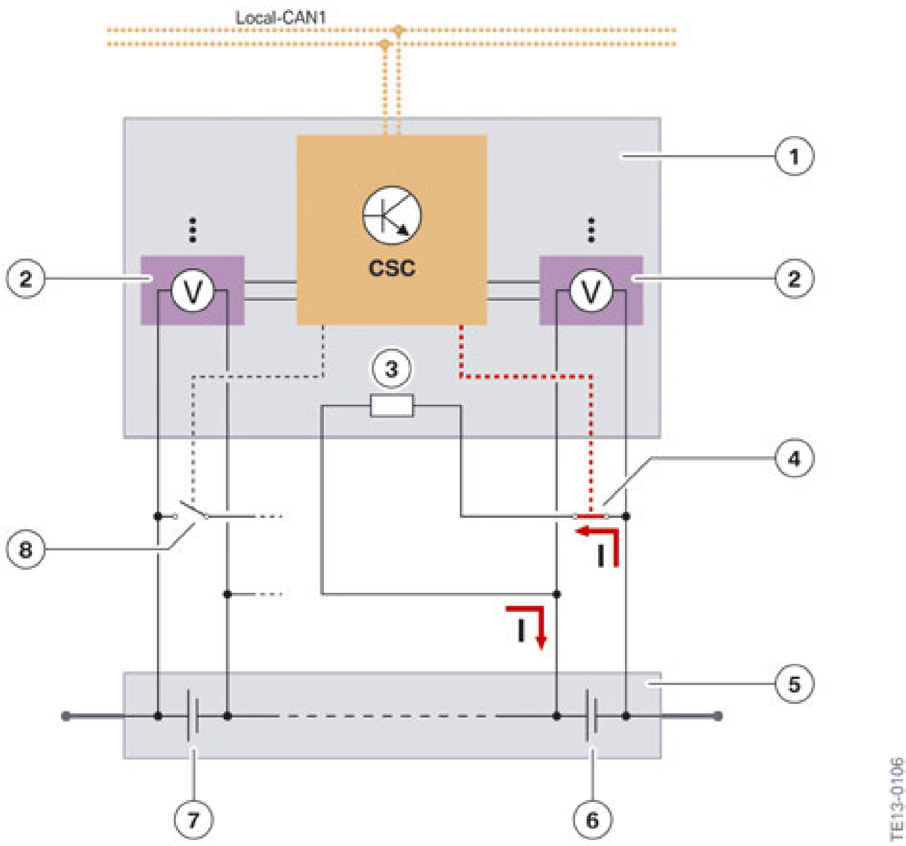
G12 PHEV principle wiring diagram of the local CANs in the high-voltage battery unit

Index	Explanation
1	Terminating resistor of Local CAN1 in the SME control unit
2	Terminating resistor of Local CAN1 in the SME control unit
3	Terminating resistor of Local CAN2 in the SME control unit
4	Terminating resistor of Local CAN2 in the safety box
5	High-voltage battery unit

If the resistance is measured at the Local Controller Area Networks as part of troubleshooting, a value of approx. 60 Ω is obtained if all bus users are connected and the termination is OK.

If one or several of the battery cells has a significantly lower cell voltage than all other battery cells, the usable energy content of the high-voltage battery is restricted. The end of energy consumption for the discharging procedure is in fact determined by the "weakest" battery cell: If the voltage of the weakest cell has fallen to the discharge limit, the discharging procedure must be terminated, also if the other battery cells still have enough energy stored. If the discharging procedure is continued however, the weakest battery cell may incur permanent damage. For this reason there is a function to adjust the cell voltage to approximately the same level. This process is called "cell balancing".

The SME control unit compares all cell voltages. The battery cells with a significantly higher cell voltage than the other battery cells are discharged during this procedure. Discharging is started by the SME control with a request via the local CAN1 to the respective Cell Supervision Circuits for these battery cells. Each Cell Supervision Circuit has an ohmic resistance for each battery cell, via which the discharge current can flow as soon as the respective electronic contact has been closed. After the discharging procedure is started it is performed or continued independently by the Cell Supervision Circuits, also if the main control units have switched to rest state in the meantime. This is made possible by the fact that the Cell Supervision Circuit control units obtain their voltage supply from the battery management electronics, which is connected directly to terminal 30. The discharging procedure is automatically ended if the voltage of all battery cells is in a specified narrow range. The cell balancing is continued until all cells have the same voltage.



G12 PHEV, principle wiring diagram: Adjusting the cell voltages

Index	Explanation
1	Cell Supervision Circuit
2	Sensors for measuring the cell voltage
3	Discharge resistor
4	Closed (active) contact for discharging a battery cell
5	Cell module
6	Battery cell whose cell voltage is reduced by discharging
7	Battery cell not being discharged
8	Open (inactive) contact for discharging a battery cell

The adjustment of the cell voltages is a procedure which involves losses. However, the electrical energy lost is very low (less than 0.1 % SoC). In contrast, the advantages are that the attainable range and the service life of the high-voltage battery are maximized which makes sense overall and is necessary for balancing the cell voltages. This procedure can naturally be performed when the vehicle is at a standstill.

2. High-voltage Battery Unit

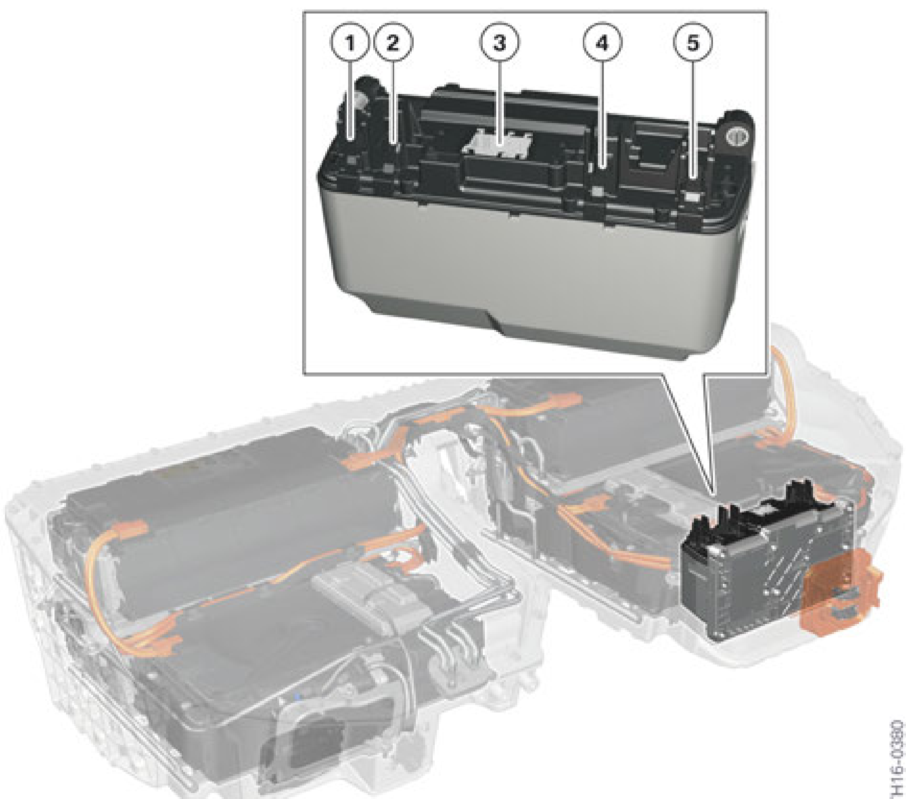
The conditions for adjusting the cell voltages are as follows:

- The motor vehicle is in the vehicle condition PARK,
- AND high-voltage system shut down,
- AND deviation of the cell voltages or the individual States of Charge of the cells are greater than a threshold value,
- AND total SoC of the high-voltage battery is greater than a threshold value.

The adjustment of the cell voltages is effected automatically when the specified conditions are fulfilled. The customer therefore does not receive a Check Control message nor does he have to implement a special measure. Also after the replacement of cell modules the SME control unit automatically identifies any necessary requirement to adjust cell voltages.

If the cell voltages however have too great a deviation or the adjustment of the cell voltage is not successful, a fault code entry must be created in the battery management electronics control unit. The customer is made aware of this fault status by a Check Control message. With help of the diagnosis system, the fault memory must then be evaluated and corresponding repair work carried out.

Safety box



G12 PHEV, installation location of the safety box

Index	Explanation
1	Positive to high-voltage connection
2	Negative from cell module 1
3	Connection, communication wiring harness
4	Positive from cell module 6
5	Positive to high-voltage connection

In each high-voltage battery unit there is an interface unit with its own housing, which is also called a "safety box" or "S-box" for short.

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The safety box is located inside each high-voltage battery unit. For this reason, an exchange is only permissible for qualified technicians.

The following components are integrated in the safety box:

- Current sensor in the current path of the negative battery terminal
- Safety fuse in the current path of the positive battery terminal
- Two electromechanical switch contactors (one switch contactor per current path)
- Pre-charge switch for slow start-up of the high-voltage system
- Voltage sensors for monitoring the switch contacts, measuring the overall battery voltage as well as monitoring the isolation resistance
- Electrical wiring for isolation monitoring

Wiring harnesses

There are two wiring harnesses in the high-voltage battery unit.

- Communication wiring harness for the connection of the Cell Supervision Circuits to the SME control unit.
- Communication wiring harness for connection of the SME and the safety box with the interface to the 12 V vehicle electrical system.

No repairs can be carried out on the wiring harness. If the connection between the cable and connector is faulty or loose, the entire wiring harness must be replaced.

2.4.2. Mechanical components

The mechanical components of the high-voltage battery unit include:

- The upper and lower housing sections
- The gasket between the upper and lower housing sections
- The four evaporators
- The venting unit
- The two module intermediate flooring panels
- The brackets for the SME and safety box

2.5. Functions

In the G12 PHEV **central functions of the high-voltage system** are controlled and coordinated by the Electrical Machine Electronics (EME). They are:

- Voltage conversion from direct current voltage in to three-phase AC voltage (electric motor mode)
- Voltage conversion from three-phase AC voltage to direct current voltage (charging mode)
- Voltage conversion from high voltage to low voltage (12 V battery charge)
- High-voltage power management
- Actuation of 12-V actuators
- Discharging the link capacitors

The high-voltage battery unit and the SME control unit are of decisive importance for the central functions of the high-voltage system. They are:

- Starting
- Regular shutdown
- Quick shutdown
- Battery management
- Charging the high-voltage battery
- Monitoring functions

2.5.1. Starting

The sequence for starting the high-voltage system is always the same irrespective of which of the following events was the trigger:

- The vehicle is “woken up” or driving readiness is established.
- Charging the high-voltage battery should start.
- Preparation of the vehicle for the journey (climate control of the high-voltage battery or the passenger compartment).

The individual steps for starting the high-voltage system are:

1 The EME control unit requests start-up via bus signals PT-CAN / PT-CAN2 / PT-CAN3.
2 The high-voltage system is monitored using self-diagnosis functions.
3 The voltage in the high-voltage system is increased continuously via the precharging circuit.
4 The contacts of the switch contactors are fully closed.

The high-voltage system is mainly monitored by the Electrical Machine Electronics control unit and the battery management electronics control unit. Criteria relevant for safety, for example the circuit of the high-voltage interlock loop or the isolation resistance, are checked. Functional preconditions such as the operating readiness of all subsystems must also be fulfilled for starting.

The high-voltage system features capacitors with high capacity values (link capacitors in the power electronics) and for this reason, it is not permitted to simply close the contacts of the electromechanical switch contactor. Extremely high current pulses would damage both the high-voltage battery and the link capacitors and the contacts of the switch contactors. First of all, the switch contactor at the negative terminal is closed. There is a precharging circuit with a resistor parallel to the switch contactor at the positive terminal. This is now activated and a switch-on current restricted by the resistor charges the link capacitors. When the voltage of the link capacitors has reached the approximate value of the battery voltage, the precharging circuit is opened and the switch contactor at the positive terminal of the high-voltage battery unit is closed. The high-voltage system is now fully operational.

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The consecutive closing of the switch contactors during starting is audible in the vehicle and does not indicate a malfunction.

If there is no fault in the high-voltage system, the entire starting of the high-voltage system is completed in approx. 0.5 seconds.

The SME control unit communicates successful starting via the PT-CAN3 to the EME control unit. Fault statuses are also communicated in the same way, if, for example, a contact of a switch contactor was unable to be closed.

2.5.2. Regular shutdown

When it comes to shutting off the high-voltage system a distinction is made between regular shut-off and fast shut-off. The regular shutdown described here protects all respective components on the one hand, and, on the other hand, includes the monitoring of components of the high-voltage system which are relevant for safety.

If the following preconditions or criteria are present, the high-voltage system is shutdown in the regular manner:

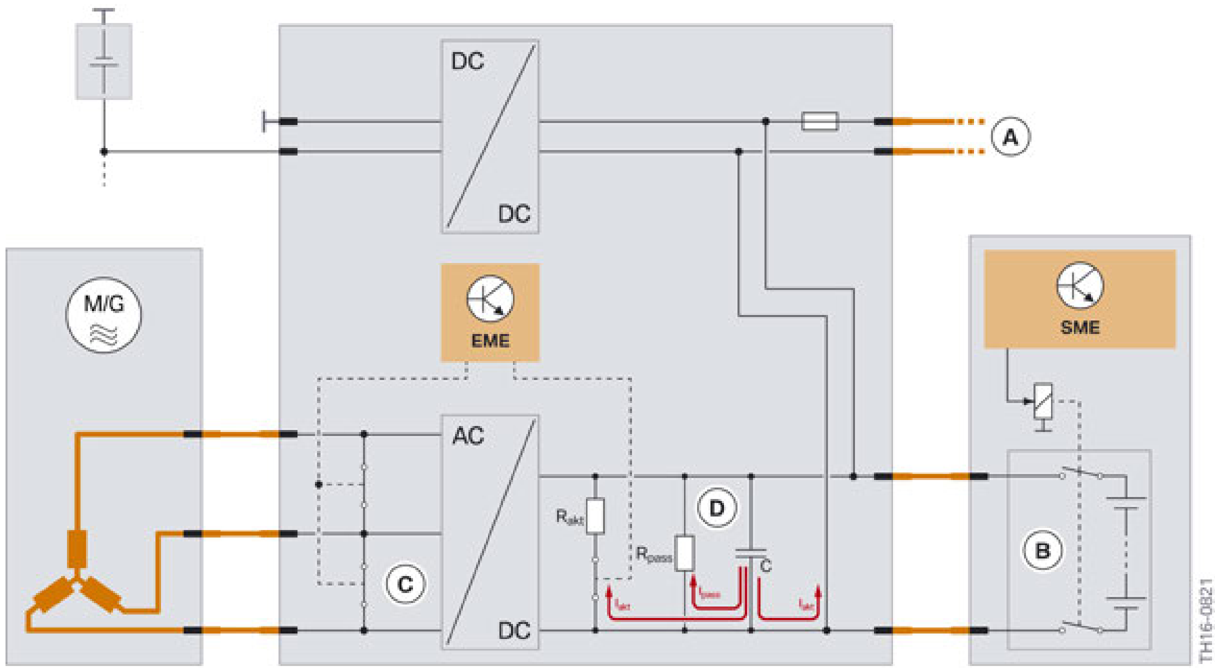
- The driver establishes vehicle condition PARK or RESIDE (terminal 15 switched off, and the after-running period has expired, controlled by the EME).
- End of the functions stationary cooling, auxiliary heater or conditioning of the high-voltage battery unit.
- End of the charging procedure for the high-voltage battery unit.
- End of the charging procedure for the system battery.
- Programming procedure of a high-voltage control unit.

The sequence for the regular shutdown is generally the same irrespective of the trigger. The individual steps are:

1. After the after-running period has expired, the EME commands a shutdown via bus signals on the PT-CAN/PT-CAN2/PT-CAN3.
2. The systems at the high-voltage electrical system (EME, KLE, EKK, EH) reduce the currents in the high-voltage electrical system to zero.
3. The coils of the electrical machine are short-circuited.
4. Opening the switch contactors in the high-voltage battery unit (controlled by SME).
5. Checking the high-voltage system, e.g. as to whether the contacts of the electromechanical switch contactors were correctly opened.

The high-voltage system is discharged, i.e. active discharge of the link capacitors (EME):

1. First, it is attempted to supply the stored energy to the 12 V system battery.
2. If this is not possible, the link capacitors are discharged via actuatable resistors.
3. If the link capacitors have not been discharged below a voltage of 60 V within 5 s, they are discharged via passive resistors.



G12 PHEV Schematic diagram of regular shutdown

Index	Explanation
A	Switch-off of all high-voltage components
B	Opening of switch contactors
C	Short-circuit of the coils of the electrical machine
D	Discharge of the link capacitors

Discharge of the link capacitors takes place in several stages as necessary.

Both the transition to the vehicle condition PARK and the shutdown can take a number of minutes. The automatic monitoring functions are a reason for this, for example. The regular shutdown is interrupted if in the meantime either a request for a renewed start-up is made or a condition has arisen to request a quick shutdown.

2.5.3. Quick shutdown

The overriding aim here is to shut down the high-voltage system as quickly as possible. This quick shutdown is then always carried out if for safety reasons the voltage in the high-voltage system has to be reduced to a safe value as quickly as possible. The following list describes the triggering conditions and the functional chain leading to the quick shutdown.

- Accident:
  Advanced Crash Safety Module (ACSM) identifies an accident. Depending on the severity of the accident, the switch-off is requested via bus signals or by a forced disconnection of the safety battery terminals from the positive terminal of both 12 V batteries. In the second scenario the voltage supply of the electromechanical switch contactors is automatically interrupted and their contacts open automatically.
- Overload current monitoring:
  With help of a current sensor in the high-voltage battery unit the current level in the high-voltage electrical system is monitored. If too high a current level is identified, the battery management electronics control unit causes a hard opening of the electromechanical switch contactor. Considerable wear occurs to the contacts of the switch contactors as a result of this opening under a high current, which must be accepted to protect the other components from damage.
- Protection in the event of a short circuit:
  In each high-voltage battery unit there is an overcurrent fuse which interrupts the high-voltage circuit in the event of a short circuit.
- Critical cell state:
  If a Cell Supervision Circuit identifies extreme undervoltage, overvoltage or excess temperature at a battery cell, this also leads to a hard opening of the electromechanical switch contactors - controlled by the battery management electronics control unit. Although this may lead again to increased wear at the contacts, this quick shutdown is necessary to prevent destroying the respective battery cells.
- Malfunction of the 12 V voltage supply of the high-voltage battery unit:
  In this case the battery management electronics control unit no longer works and it is no longer possible to monitor the battery cells. For this reason the contacts of the electromechanical switch contactors also open here automatically.

In addition to disconnecting the high-voltage system the link capacitors are also discharged (Electrical Machine Electronics) and the coils of the electrical machine (Electrical Machine Electronics, EKK) are short-circuited. The high-voltage control units receive the request on the one hand by bus signals and identify this condition on the other hand by the sudden drop in the current level in the high-voltage circuit.

2.5.4. Charging

The SME control unit also plays an important role when charging the high-voltage battery, regardless of whether it is by energy recovery, raising the load point of the combustion engine or from the external power supply system. Using the State of Charge and the temperature of the battery cells, the battery management electronics control unit determines the maximum electrical power the high-voltage battery unit can currently absorb. This value is transmitted in the form of a bus signal via the PT-CAN3 to the EME control unit. The high-voltage power management function coordinates the individual power requirements.

During charging the SME control unit constantly identifies the State of Charge already reached and monitors all sensor signals of the high-voltage battery. In order to ensure optimal progress of the charging procedure, the SME control unit constantly calculates current values for the maximum charging power based on these values and communicates these to the EME control unit. The cooling system of the high-voltage battery unit is continuously controlled by the SME control unit during the charging procedure. This contributes to a quick and efficient charging procedure.

To achieve the highest possible electrical range, preheating/precooling of the vehicle interior (heating or cooling) must be complete with the charging cable connected. Consequently the required energy is not drawn from the high-voltage battery unit, but supplied directly from the convenience charging electronics.

Further details about the charging procedure, in particular providing the power supply from a public power grid up to the convenience charging electronics KLE in the G12 PHEV are described in the "G12 PHEV High-voltage Components" Training Reference Manual.

2.5.5. Monitoring functions

12 V supply voltage from the safety battery terminal

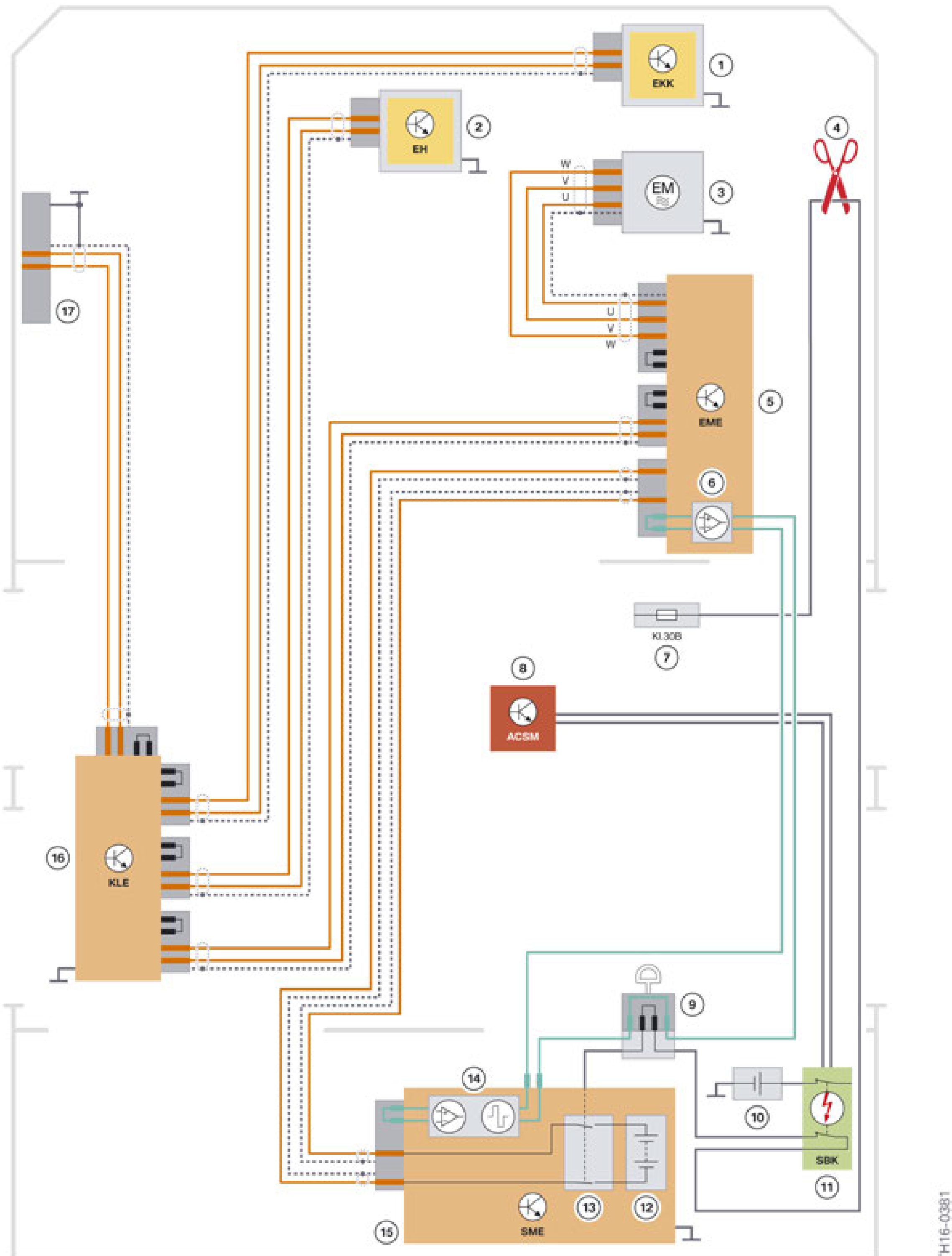
To be able to perform a quick shutdown of the high-voltage system in the event of an accident of corresponding severity, the solenoids of all electromechanical switch contactors are supplied with 12 V from the safety battery terminal. If the safety battery terminal is disconnected in the event of an accident, this supply voltage is no longer necessary and the contacts of the switch contactors open automatically.

In addition, the battery management electronics control unit evaluates the voltage on this line electronically and also causes the high-voltage system to shut down including discharge of the link capacitors and the active short circuit of the electrical machine.

High voltage interlock loop

The SME control unit evaluates the signal of the high-voltage interlock loop and checks whether there is an open circuit within this circuit. In the event of an open circuit, the battery management electronics control unit can cause the high-voltage system to perform a quick shutdown.

2. High-voltage Battery Unit



G12 PHEV, system wiring diagram of high-voltage interlock loop

Index	Explanation
1	Electric A/C compressor
2	Electrical Heating (EH)
3	Electric motor
4	Rescue disconnect
5	Electric Motor Electronics (EME)
6	Evaluation circuit for test signal of the high-voltage interlock loop in the Electrical Machine Electronics
7	Power distribution box, front left
8	Advanced Crash Safety Module (ACSM)
9	High-voltage safety connector ("Service Disconnect")
10	Battery
11	Safety battery terminal (SBK)
12	Cell modules
13	Switch contactors
14	Evaluation circuit for test signal of the high-voltage interlock loop in the battery management electronics
15	Battery management electronics (SME)
16	Convenience charging electronics (KLE)
17	Charging socket

The electronics for controlling and generating the test signal for the high-voltage interlock loop are integrated in the G12 PHEV in the battery management electronics (SME). Generating the test signal starts when the high-voltage system is to be started and ends when the high-voltage system has been shut down. A square-wave AC signal with a frequency of approx. 88 Hz is generated as the test signal by the battery management electronics and fed into the test lead. The test lead has a ring configuration (similar to that of the MOST bus). The signal of the test lead is evaluated at two points in the ring: in the Electrical Machine Electronics and finally right at the end of the ring, in the battery management electronics. If the signal is outside a fixed, defined range, a disconnection of the circuit or a short circuit to vehicle ground is identified in the test lead and the high-voltage system is switched off immediately. If the high-voltage interlock loop at the high-voltage safety connector (“Service Disconnect”) is disconnected, then the switch contactors are opened directly. In addition, all high-voltage components are switched off.

Contacts of the switch contactors

After the battery management electronics control unit has requested the contacts of the switch contactors to open during the shutdown of the high-voltage system, with help of a voltage measurement a check parallel to the contacts checks whether they have actually been opened. In the highly unlikely case that the contact of a switch contactor does not open, there is no direct danger for the customer or the Service technician. However, for safety reasons a renewed start-up of the high-voltage system is prevented. It is no longer possible to continue the journey using the electric motor.

2. High-voltage Battery Unit

Pre-charge switch

If for example during the start-up of the high-voltage system a fault is identified with the pre-charge switch, then the start-up is cancelled immediately and the high-voltage system is not put into operation.

Excess temperature

The cooling system of the high-voltage battery unit ensures that the temperature of the battery cells is in the optimal range in all operating conditions. If due to a fault the temperature of one or several battery cells increases to the extent that the optimal range is left, the power is reduced initially to protect the battery cells. If the temperature continues to increase and thus threatens damage to the battery cells, the high-voltage system is switched off in good time.

Undervoltage

Undervoltage at a battery cell is avoided by the constant monitoring and adjustment of the cell voltage as required. The total voltage of the entire high-voltage battery unit is also monitored and used to determine the State of Charge. If the total voltage has fallen to the extent that the high-voltage battery is at risk of a total discharge, a further discharge is prevented. It is no longer possible to continue the journey using the electric motor.

Isolation monitoring

The insulation monitoring determines whether the insulation resistance between active high-voltage components (e.g. high-voltage cables) and earth is above or below a required minimum value. If the insulation resistance falls below the minimum value, the danger exists that the vehicle parts will be energized with hazardous voltage. If a person were to touch a second active high-voltage component, he or she would be at risk of electric shock. This is why there is fully automatic insulation monitoring for the high-voltage system of the G12 PHEV.

In contrast to the high-voltage battery units of the BMW i vehicles, the insulation monitoring is now located in the safety box. This has the advantage that it is no longer necessary to route high-voltage cables to the battery management electronics. The safety box sends results to the SME control unit via local CAN2, and evaluates the results of the measurements. When the high-voltage system is active, the safety box monitors isolation at regular intervals (approx. every 5 s) by measuring the resistance (indirect isolation monitoring). Earth serves as the reference potential.

Without additional measures only local insulation faults in the high-voltage battery unit could be determined in this way. However, it is equally important to identify insulation faults from the high-voltage cables in the vehicle to earth. For this reason all the electrically conductive housings of high-voltage components are conductively connected to earth. This enables isolation faults in the entire high-voltage electrical system to be identified, from a central point in fact, the high-voltage battery unit.

The insulation monitoring responds in two stages. When the insulation resistance drops below a first threshold value, there is still no direct danger to people. The high-voltage system therefore remains active; no Check Control message is output, but the fault status is naturally stored in the fault memory. In this way the Service technician is alerted the next time the car is in the workshop and can then check the high-voltage system. When the insulation resistance drops below a second, lower threshold value, this is accompanied not only by the storage of the fault in the fault memory, but also by the appearance of a Check Control message prompting the driver to visit a workshop.

As there is no direct danger for the customer or Service technician by such an isolation fault, the high-voltage system remains active and the customer is able to continue the journey. Nevertheless, the high-voltage system should be checked by BMW Service as soon as possible. In order to identify the component in the high-voltage system which caused the isolation fault the fault must be narrowed down by the Service technician. However, the Service technician does not have to perform a fundamental measurement of the isolation resistance himself – this task is performed by the high-voltage system through the insulation monitoring. When an insulation fault is detected, the Service technician must run through a test schedule in the diagnosis system to find the actual location of the insulation fault.

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The proper electrical connection of all high-voltage component housings to ground is an important prerequisite for ensuring that the insulation monitoring functions properly. Accordingly, this electrical connection must be restored carefully if it has been interrupted during repair work.

2.6. Service information

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Before working on high-voltage components of the G12 PHEV, it is essential to observe and implement the electrical safety rules:

1 The high-voltage system must be disconnected from the supply.

2 The high-voltage system must be secured against restart.

3 The safe isolation of the high-voltage system must be verified

G12 PHEV High-voltage Battery Unit

3. Repair

3.1. Pre-Requirements

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The following description of the repair of the high-voltage battery unit is only a general list of the content and the procedure. **In general, only the specifications and instructions in the current valid edition of the repair instructions apply.**

3.1.1. Organizational matters

Some organizational prerequisites must be fulfilled so that a repair to the high-voltage battery unit is allowed and can be adequately performed. These prerequisites concern both the dealership and the Service technician.

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Repair of high-voltage battery units is permitted only in outlets with suitably qualified workshop personnel.

The dealership must also make available the necessary special tools and a suitable workbay for the repair. The most important special tools are:

- Two-post vehicle hoist
- Mobile MHT 1200 assembly lifting table with adapters to fix the high-voltage battery unit
- Charger for cell modules of the high-voltage battery unit
- EoS test device for checking the repaired high-voltage battery unit
- Lifting device number 2 360 072 for the installation and removal of cell modules
- Panel wedge made from plastic, number 2 298 505, for removing the clips in the high-voltage battery unit
- Special tool to attach cable straps
- Barrier tape
- Yellow warnings with flash labels are recommended

The workbay for the repair of the high-voltage battery unit must be clean, dry and free of flying sparks. Therefore, avoid direct proximity to vehicle cleaning areas or work stations at which repair work to the body is performed. If necessary, use partition walls and barrier tape to separate areas.

The Service technician performing the repair of the high-voltage battery unit must also satisfy key prerequisites:

1 Qualifications
2 Strict use of the diagnosis system and the special tools
3 Follow repair instructions exactly

Only Service technicians qualified in the repair of the high-voltage battery unit can perform this work. This includes successful completion of the training “Expert in working on high-voltage intrinsically safe vehicles”, training on the high-voltage system of the G12 PHEV, and specifically on repairing the high-voltage battery unit of the G12 PHEV, or another Generation 3.0 high-voltage battery.

The diagnosis system must be used for troubleshooting before the removal and opening of the high-voltage battery unit. Only if the test schedule specifies it and the prerequisite "no external mechanical damage" is fulfilled can the high-voltage battery unit be opened and the component identified as faulty by the test schedule replaced.

Apart from the replacement of faulty components, no repair work in the high-voltage battery unit is allowed. For example, a faulty wiring harness cannot be repaired, it can only be replaced by a new one. To carry out a replacement of a faulty component, it is extremely important the operations specified in the repair instructions are followed exactly. It is also very important to use the prescribed special tools. When the Service technician satisfies all these prerequisites he can safely perform the repair of the high-voltage battery unit and deliver the high quality expected.

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Before starting work on high-voltage vehicles that have been involved in an accident, the instructions and notices in the following documents in the repair instructions must be observed:

- Safety Information for handling electric vehicles.
- Assessment of vehicle that has been involved in an accident.
- Visual inspection of high-voltage battery unit after accident.

3.1.2. Safety rules

- The workbay for high-voltage battery unit repair must be clean (no grease, dirt or swarf), dry (no escaping fluids) and there must be no flying sparks (not located in the vicinity of body repair work). Therefore, avoid direct proximity to vehicle cleaning areas (wash systems) or work stations at which repair work to the body is performed. If necessary, use partition walls to separate areas.
- Barrier tape is necessary to protect the workbay against unauthorized access (insufficient qualification, customers, visitors, etc.), as well as in the event of a lack of intrinsic high-voltage safety or unclear condition of the component. It is recommended to position yellow warnings with a lightening flash when the working area is left unattended.
- Residual moisture and coarse dirt contamination on the upper housing section of the high-voltage battery unit must be removed before disassembly.
- A careful visual inspection of the components being processed must be performed before and after each operation. For example, during the removal of a component other components which become loose as a result of the removal must be checked for damage.

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  If the housing or the internal high-voltage components are damaged, repairs may exclusively be performed by BMW qualified electrician, contact Technical Support PuMA as a mandatory requirement. **Without this qualification, work on the high-voltage battery unit must be terminated immediately for safety reasons.**

3. Repair

- The first step after the housing cover is opened for the repair of the high-voltage battery unit is a visual inspection for mechanical damage.
- Before working in the open high-voltage battery unit, always disconnect the high-voltage cable between cell modules 3 and 4 in order to interrupt the series connection.
- In the event of work interruptions attach the removed housing cover again and secure against unintentional opening by screwing in some screws. Position yellow warning cones on the housing cover and cordon off the working area using barrier tape.
- The use of pointed or sharp-edged tools or similar objects is prohibited for the entire internal repair of the high-voltage battery unit. For example, screwdrivers, diagonal cutting pliers, knives, etc. are prohibited. Plastic panel wedges, socket wrenches, flat-nose pliers, and approved special tools are permissible.
- Open cable straps on the low-voltage wiring harness using combination pliers only. Cable straps on the high-voltage cables may only be opened using combination pliers in order to disassemble the high-voltage cable including the bracket. Use the appropriate special tools to install cable straps.
- During the removal and installation of the cell modules ensure when slackening and removing the screws that the plastic cover of the cell modules is not removed. The live cell contact system is located underneath.

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The repair instructions 6127 ... "Notes on opening and closing cable straps on electric and hybrid vehicles" must always be observed when opening and replacing cable straps.

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If the plastic cover of the cell module is loose, it must be secured. If it is not possible to secure the cover or the cover is broken, the cell module must be replaced. Assuming appropriate qualification, the individual plastic cover may also be exchanged by a **BMW qualified electrician** by creating a PuMA case.

- In the event of dirt contamination in the high-voltage battery unit, the affected areas must be cleaned thoroughly after clarification of the cause. The following cleaning agents are approved:
  - Spirit
  - Windscreen cleaning agent
  - Glass cleaning agent
  - Distilled water
  - Vacuum cleaner with plastic attachment
- Do not leave any tools inside; before closing the housing cover check the tools in the tool kit for completeness.

- In the high-voltage battery unit lost/fallen small parts/screws must be removed. To avoid losing screws during repair work on the high-voltage battery unit, it is generally recommended to use magnetic tools.
- Due to the relatively thin wall thickness of the evaporator, there is an increased risk of damage during disassembly and installation. Exercise extreme caution when handling the evaporator as cell module cooling can no longer be guaranteed if the heat exchanger is damaged (bent, dented). Consequently, the available range and power of the vehicle are reduced considerably. Seals and sealing surfaces (venting unit, high-voltage connector, 12 V connector, evaporator connection) must be cleaned with the specified cleaning agents before reassembly.
- Electrolyte:
  The majority of electrolyte is bound in the solid cathode material, primarily lithium-nickel-manganese-cobalt-oxide, and in the solid anode material, mainly graphite. The amount of the free electrolyte in the high-voltage battery unit is very small. In the event of a leak, electrolyte and solvent vapors may be released. Wash thoroughly using clear water in the event of contact with the skin or eyes and seek immediate medical advice.
  Mainly flammable gases, asphyxiating gases and hazardous substances such as carbon monoxide, carbon dioxide, hydrogen and hydrocarbons develop in the event of a fire.
  Attention! Do not breathe in! There must be adequate fresh air. If breathing stops use an artificial respiratory device and contact a doctor immediately.
  Inform the fire brigade in the event of a fire. Clear the area straight away and secure the scene of the accident. Only try to extinguish the fire if it does not pose a danger and use a suitable extinguisher, e.g. water, foam or CO_2 fire extinguisher.

3.1.3. Electrical and mechanical diagnosis

- Analysis of faults preventing removal (e.g. duplicate contactor label).
- Troubleshooting faults that do not allow an accurate statement about the internal condition of the high-voltage battery unit, (e.g. internal isolation faults).
- Determine repair measure from test schedule (diagnosis) and print out location diagram.

⚠

The fault memory can only be deleted after the high-voltage battery unit has been fully repaired.

- If cell modules are to be replaced, determine the State of Charge or the voltage of the intact cell modules.
- Use the State of Charge of a **new** cell module as the reference value when exchanging **all** cell modules. Read out: Connect charger/discharging device to a new cell module, read out State of Charge/voltage and use as nominal voltage for all other cell modules.
- Visual inspection for dirt contamination and damage to housing when installed, as well as to the connections and venting unit. Damage to the diaphragm of the venting unit may indicate damaged cells. If this is the case, special care must be taken when checking the inside of the open high-voltage battery unit.

3.1.4. Removal of the high-voltage battery unit from the vehicle

1 Evacuate refrigerant.
2 De-energize the high-voltage system via the high-voltage service disconnect; secure against switching back on, and make sure that the voltage is zero.
3 Complete the preliminary work as per the repair instructions (e.g. removing the underbody panelling, exhaust system and drive shaft).
4 Disconnect connections (12 V vehicle electrical system, high-voltage, refrigerant).
5 Close connections for refrigerant lines with plugs.
6 Position the mobile workshop crane including multi-purpose lifting tool on the high-voltage battery unit, secure it and visually check it is positioned correctly.
7 Undo the mounting bolts on the high-voltage battery unit.
8 Lever out the high-voltage battery unit.
9 Visual inspection for dirt contamination and damage to housing on all sides.
10 Check thermal abnormalities in the event of faults which may indicate unclear condition of the high-voltage battery unit.
11 Transport to the mobile lifting table as the repair workplace.

3.2. Repair of the removed high-voltage battery unit

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The following description of the repair of the high-voltage battery unit is only a general list of the content and the procedure. **In general, only the specifications and instructions in the current valid edition of the repair instructions apply.**

3.2.1. General and preliminary measures

The high-voltage battery unit is a component with large dimensions and heavy weight. Only the combination of the housing and the cell modules secured inside give the high-voltage battery unit complete stability (rigidity), as is required when driving.

For the replacement of cell modules and Cell Supervision Circuits it is extremely important to enter the serial number and the installation position of the components replaced in the print-out of the report and in the battery management electronics (SME). There is a Service function in the diagnosis system for the start-up of the high-voltage battery unit after repair. Otherwise, the new installation position is automatically assigned by the battery management electronics (SME). The results are not correct location data.

⚠

In the event of a subsequent repair of the high-voltage battery unit the diagnosed faults are displayed for the installation position and the incorrect cell module or Cell Supervision Circuit is replaced.

3.2.2. Work before opening

Prepare the workbay

- Prepare the mobile assembly lifting table with adapters
- Cleanliness of the workbay
- Keep away from emerging fluids
- No tools or other objects at the workbay
- Separation from other work stations by a separate room or using barrier tape is recommended
- No flying sparks in the vicinity, apart from that erect corresponding partition walls

3.2.3. Disassembly of the housing sections of the high-voltage battery unit

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In general, for all operations at the high-voltage battery unit **the current repair instructions** must be followed.

The following procedure is prescribed for the disassembly of the housing parts: Firstly, all dirt contamination and any moisture must be removed from the housing cover. The following cleaning agents are approved:

- Spirit
- Windscreen cleaning agent
- Glass cleaning agent
- Distilled water
- Vacuum cleaner with plastic attachment

Undo the screws of the housing cover and remove any contamination from the screw holes using a vacuum cleaner. Carefully remove the housing cover. Visually check the open high-voltage battery unit for any damage and ingress of moisture.

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Immediately stop work if damage is visible and contact a specialist BMW qualified electrician or technical support by creating a PuMA case.

⚠

The screws of the housing parts must be replaced each time after disassembly. Clean the contact surfaces of the seals on the upper housing section and the lower housing section.

3.2.4. Removal of the cell modules

⚠

The following description of the repair of the high-voltage battery unit is only a general list of the content and the procedure. **In general, only the specifications and instructions in the current valid edition of the repair instructions apply.**

Print the position plan from the diagnosis system before removing the cell modules or Cell Supervision Circuits (CSC). First and foremost, you must comply with the safety rules and disconnect the high-voltage cable between cell module three and cell module four. Now all cell modules and Cell Supervision Circuits must be numbered using a waterproof pen according to the location diagram.

Before a bottom cell module can be removed, the cell module above it must be removed, including the module intermediate flooring panel and the evaporator.

It is now possible to remove the affected cell modules. The high-voltage connectors of the cell modules must be disconnected for this. Then unlock and undo the Cell Supervision Circuit connectors. The Cell Supervision Circuits are removed together with the cell modules.

Then the mounting bolts of the cell modules can be slackened. If necessary, remove the CSC ring wiring harness. Use a panel wedge if required. Under no circumstances use any sharp objects. Carefully lift out the cell module with the special tool; while doing so, pay attention to the clearance between the cell modules. Set down the cell module with base on a clean surface and ensure it is positioned so it does not slip or tilt.

⚠

When lifting out the evaporator, hold it at the respective ends. Otherwise, there is a high risk of damage due to the low mechanical stability.

3.2.5. Preparation before the installation of a cell module

Before the installation of a new cell module its State of Charge is brought to the level of the remaining cell modules, which was read out beforehand. Clean the high-voltage battery unit, the evaporator and the cell module flooring panel prior to installing the cell module.

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If all cell modules have to be replaced, alternatively the voltage of one cell module can be used as a nominal charging voltage for all other cell modules in order to minimize the charging times (read out using charger).

3.2.6. Installation of the cell modules

Mount the Cell Supervision Circuits to the cell module prior to installing the cell modules. Carefully lift the cell module using a special tool. Pay attention to neighboring parts. Tighten the screws of the cell module to the specified torque during installation. After installing the bottom cell module, connect the Cell Supervision Circuit wiring harness connector to the Cell Supervision Circuits. Connect the high-voltage connector of the respective cell module. Before installing the module intermediate shelf, carry out a visual check to ensure that all lines and connectors are correctly routed and connected.

You can then mount the top evaporator. Carefully lift the top cell modules onto the evaporator using the special tool, and tighten the mounting bolts to the specified torque. In the next step, the wiring harness of the Cell Supervision Circuits and corresponding connectors can be locked. In a final step, connect the high-voltage cable between cell modules three and four.

Take a note of the serial number of the new cell module and its installation position in the high-voltage battery unit from the report printed out from the diagnosis system. There is a Service function in the ISTA workshop information system for the start-up of the high-voltage battery unit after repair. Here the serial numbers of the new cell modules must be entered in the battery management electronics control unit.

3.2.7. Removal of the Cell Supervision Circuits

Before the removal of the cell modules or Cell Supervision Circuits (CSC) the location diagram must be printed out from the diagnosis system. First and foremost, you must comply with the safety rules and disconnect the high-voltage cable between cell modules three and four. Now all cell modules and Cell Supervision Circuits must be numbered using a waterproof pen according to the location diagram. Detach the Cell Supervision Circuit ring wiring harness over a wide area. Use a panel wedge for this if required. Under no circumstances use any sharp objects. Disconnect and remove the connectors of the upper Cell Supervision Circuit. Remove the cell modules in order to remove the bottom Cell Supervision Circuit. The Cell Supervision Circuit can then be unclipped.

3.2.8. Installation of the Cell Supervision Circuits

Note down the serial number of the new Cell Supervision Circuit and its installation position in the high-voltage battery unit on the slip printed out from the diagnosis system. Connect and attach the Cell Supervision Circuits to the cell module. The components previously disassembled are installed in reverse order. In a final step, connect the high-voltage cable between cell modules three and four. There is a Service function in the ISTA workshop information system for the start-up of the high-voltage battery unit after repair. Here the serial numbers of the new cell modules must be entered in the battery management electronics control unit.

⚠

Special features when replacing several Cell Supervision Circuits:

The documentation and entry of serial numbers and the installation position is extremely important. Otherwise, the installation position is automatically assigned. If an incorrect installation position is entered, all faults reported for a later defect of the Cell Supervision Circuit are displayed for this incorrect installation position and as a result the Cell Supervision Circuit is replaced at the wrong location!

3.2.9. Removing the cooling grids

When removing the top cooling grids, the top cell modules first need to be disassembled. When removing the bottom cooling grids, the respective module intermediate flooring panel, the respective bottom cell modules, and the Cell Supervision Circuit ring wiring harness additionally need to be disassembled.

⚠

When lifting out the cooling grids, hold it at the respective ends. Otherwise, there is a high risk of damage due to the low mechanical stability.

3.2.10. Installing the cooling grids

If the cooling grids are only removed and installed, the sealing rings on the radiator connection must be replaced. Before installing the bottom cooling grids, the plastic pin must be lined up as per the repair instructions. A faulty plastic pin must be replaced.

⚠

When installing a new cooling grid, hold the cooling grid at both ends. Otherwise, there is a high risk of damage due to the low mechanical stability.

After installing the bottom cooling grid, carefully lift in the bottom cell modules using the special tool. It must be ensured that the Cell Supervision Circuit is mounted on the cell module. Position the cell module screws and tighten to the specified torque. After installing the bottom cell module, connect the Cell Supervision Circuit wiring harness connectors to the Cell Supervision Circuit. Connect the high-voltage connectors of the respective cell modules. Before installing the top cooling grids, attach the module intermediate flooring panels.

You can then mount two top cooling grids. Carefully lift the top cell modules onto the cooling grids using the special tool, and tighten the mounting bolts to the specified torque. In the next step, you can fasten the Cell Supervision Circuits on the cell modules and lock the matching connectors. In a final step, connect the high-voltage cable between cell modules three and four.

3.2.11. Installation of the housing cover of the high-voltage battery unit

Check the sealing surface of the housing cover and lower housing section and clean if necessary. The gasket of the housing sections must be replaced. Before the housing cover is mounted, check whether the high-voltage cable is connected between cell modules three and four. Ensure that the seal does not come into contact with sharp edges when installing the housing cover.

⚠

The screws are self-tapping, therefore they must be manually positioned before work is continued using the tool. Otherwise, there is a risk of damage to the thread of the lower housing section. Use of power screwdrivers is not allowed as otherwise the screws/threads may tear. The screws must be replaced.

Replace any screw connections where the screw does not reach the specified torque or if the thread is damaged. A minimum of one screw connection must be undamaged and it must not be necessary to rework this screw connection. All other screw connections can be reworked. The upper housing section must be in contact with the high-voltage battery unit to rework the threads to prevent contamination by swarf. Replace the lower housing section if all screw connections have been damaged.

G12 PHEV High-voltage Battery Unit
3. Repair

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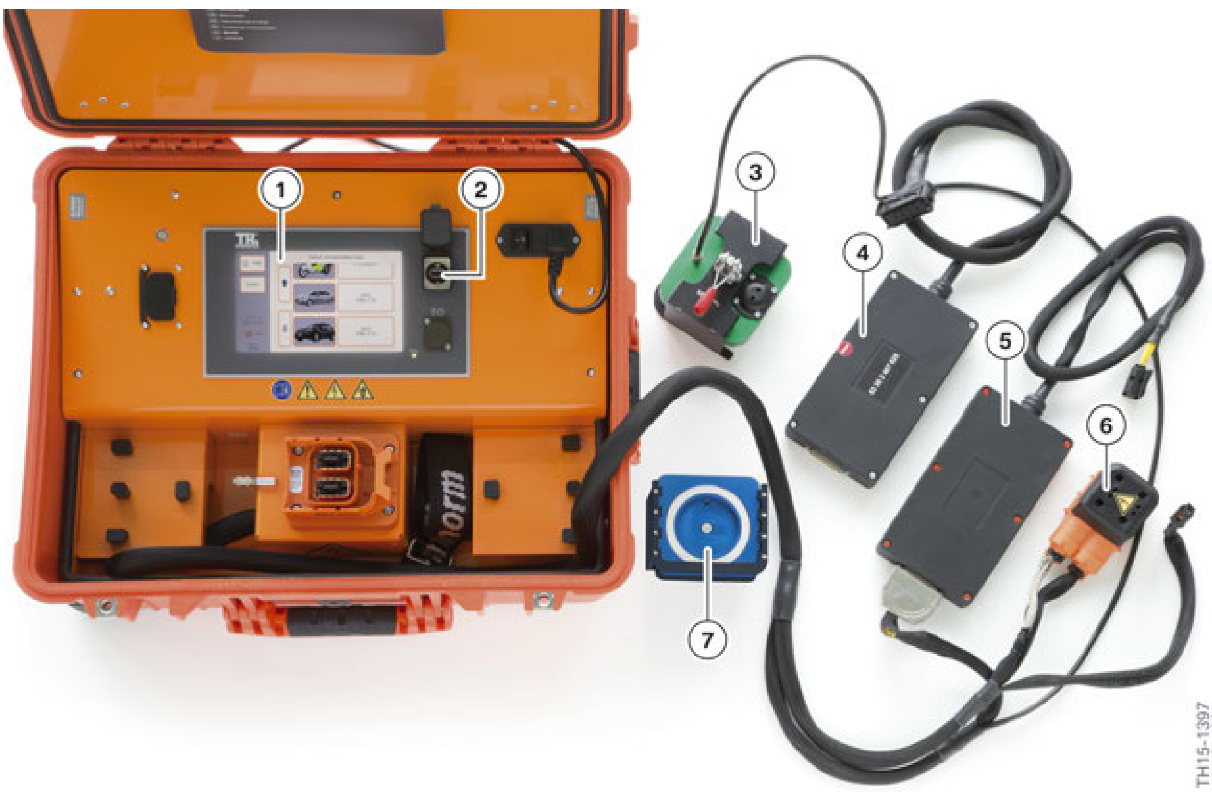
In the event of repairs on a damaged screw thread in the lower housing section the housing bore must be restored using a Kerb Konus insert. Any swarf must be removed from the high-voltage battery unit.

Tighten the new screws to the specified torque.

3.3. Final work

3.3.1. Final test with EoS tester



EoS tester

Index	Explanation
1	Touch screen for operation
2	USB interface for updates
3	Pressure bell BMW i3
4	Relay box for high-voltage test PHEV
5	Relay box for BMW i high-voltage test
6	High-voltage connector
7	Pressure bell PHEV/BMW i8

Before installation, a test must be performed with the EoS tester. Mount the appropriate test adapter for the venting unit. Connect the test connections for the pressure connection, high-voltage connector and interface to the 12 V on-board mains connector.

To start the complete test, it is first necessary to select the corresponding vehicle and/or the high-voltage battery unit. Then the tests for de-energized state, isolation resistance and SME isolation monitoring are carried out. Now the fault memory is read and if there are no faults a test code is issued.

If a fault is established in the leakage test, the test will not be aborted immediately. All electrical checks will be performed first and then the result presented. If the electrical checks are OK, a switch-over to leakage detection mode occurs. If the electrical check also fails, a message is displayed with a request to contact technical support.

Read out the test code and take a note of it for subsequent transfer to the diagnosis system or scan the Data Matrix Code (DMC) directly. Finally, the EoS tester sets the transport bit so that the switch contactors cannot be activated. During final diagnosis the service function for high-voltage battery start-up resets the transport bit after having entered the test code to activate the switch contactors and transfer FASTA data to the workshop system.

3.3.2. Installation of the high-voltage battery unit in the vehicle

Carefully insert the high-voltage battery unit into the vehicle using the mobile assembly lifting table. Make sure the high-voltage battery unit is correctly locked and centred during lifting. Slowly lift the high-voltage battery unit and align the mounting brackets. Position the mounting bolts and completely lift the high-voltage battery unit. Tighten the mounting bolts to the specified torque.

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The exact procedure must be followed when installing the potential compensation bonding screws:

1 Clean contact surfaces and have checked by a second person.
2 Tighten the mounting bolts to the specified torque.
3 Have the torque checked by a second person.
4 Both persons must record this in the vehicle records for the correctness of the version.

The O-rings must be replaced before mounting the heating and air-conditioning cables. Then the connectors for the high-voltage cables and the interface for the 12 V vehicle electrical system lines must be connected.

3.3.3. Final electrical diagnosis

In the diagnosis system the service function for the start-up of the high-voltage battery unit must be started. Enter the test code of the EoS diagnosis system or scan via DMC. The serial numbers and the installation positions of the replaced components are transferred by the battery management electronics to the diagnosis system and documented in vehicle operation and service data transfer and analysis. The switch contactors are enabled by the diagnosis system. Then the fault memory is read. Finally, the high-voltage battery unit is fully charged and the heating and air-conditioning system is filled.

4. Disposal

4.1. Storage of damaged batteries

A battery is considered damaged if:

- The high-voltage battery unit has visible sings of overheating (locally or otherwise).
- There is smoke or gasses emerging from the high-voltage battery unit.
- The high-voltage battery unit has a deformed or torn outer skin.

As soon as the defective batteries is identified:

- Technical service should be contacted and PuMA case should be created.
- If the vehicle has a damaged high-voltage battery it must be parked outside (for at least 48 hours) or until further recommendations are provided via a PuMA case.
- The disabled vehicle should be parked at a predetermined storage location which must be at least 5 meters away from buildings, vehicles or other combustible materials, for example waste containers.

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Only especially trained and certified BMW Technical Service personal should ever remove and handle externally damaged HV batteries. BMW dealer technicians are not permitted to remove nor handle high voltage batteries with external damage. If a HV battery is found to be externally damaged the vehicle should be isolated and Technical Service (electromobility) should be notified in the way of a PuMA case. Technical Service will then instruct the technician as to how to proceed.

4.2. Establishing suitability for transportation

The analysis and assessment of the high-voltage battery unit must be performed by a person properly trained by BMW Technical Training. The certification of suitability for transportation is the responsibility of the qualified person.

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To establish the suitability for transportation the repair instruction 6125... "Assessing the suitability for transportation of the high-voltage battery unit in Service workshops" must be followed and the appropriate form filled out and placed in vehicle file.

Establishing the suitability for transportation is divided into two areas:

- Electrical assessment
- Visual assessment

4.2.1. Electrical assessment

For checking the suitability for transportation the procedure for the cell modules has to be completed first via the diagnosis system, during which the electrical assessment of the suitability for transportation is established. The following information is available as a result:

1 The evaluation of the fault code entries indicates that the high-voltage battery unit does **not have any faults preventing transportation.** For further steps please refer to the repair instructions "Assessing the suitability of transportation of the high-voltage battery unit in Service workshops". The serial numbers of the high-voltage battery unit and the SME control unit which must be noted are also displayed.

2 The evaluation of the fault code entries indicates that the high-voltage battery unit **cannot be transported** without further measures. The causes are the following stored and current fault code entries:

- 21F016 high-voltage battery unit, voltage supply of the evaluation electronics for cell module X, short circuit to ground.
- 21F080 high-voltage battery unit, cell module X, over voltage of at least one cell. For further steps please refer to the repair instructions "Assessing the suitability of transportation of the high-voltage battery unit in Service workshops".

4.2.2. Visual assessment

The following criteria must be checked during the visual assessment:

- Smoke
- Visual localized overheating
- Areas hot to the touch
- Crack or opening on the housing
- Dents in the housing or deformation
- Heavy corrosion
- Loose or damaged connections
- Serial number/label of safety label illegible
- Suspected water damage

If one or more points are answered with yes, the further course of action must be clarified with Technical Support and a PuMA case started!

If the vehicle battery fails the Electrical or Visual assessment, apply the following; the removed high-voltage battery unit must be isolated using barrier tape. If the high-voltage battery unit is still in the vehicle the vehicle must be isolated, locked and restricted as previously explained.

4. Disposal

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Repair of the high-voltage battery is only allowed in a retail service center that has qualified and certified service technicians. These technicians must have completed the ST1507 F15 PHEV Complete Vehicle instructor led course and successfully passed the hands-on certification.

4.3. Disposal of high-voltage battery unit

If a high-voltage battery unit must be disposed of, the specified disposal company should be contacted to arrange pickup and recycling of battery. In the event of non-defective batteries the transport packaging for spare parts can be used. **Please refer to the latest BMW parts bulletin for the most current information regarding the distribution and recycling of high voltage batteries.**

In the event of a damaged battery, standard transport packaging may not be used. Contact Technical Service for further support.

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For more information regarding the appropriate procedure and guidelines on how to handle defective or damaged high voltage batteries, please refer to the latest BMW parts bulletin.

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